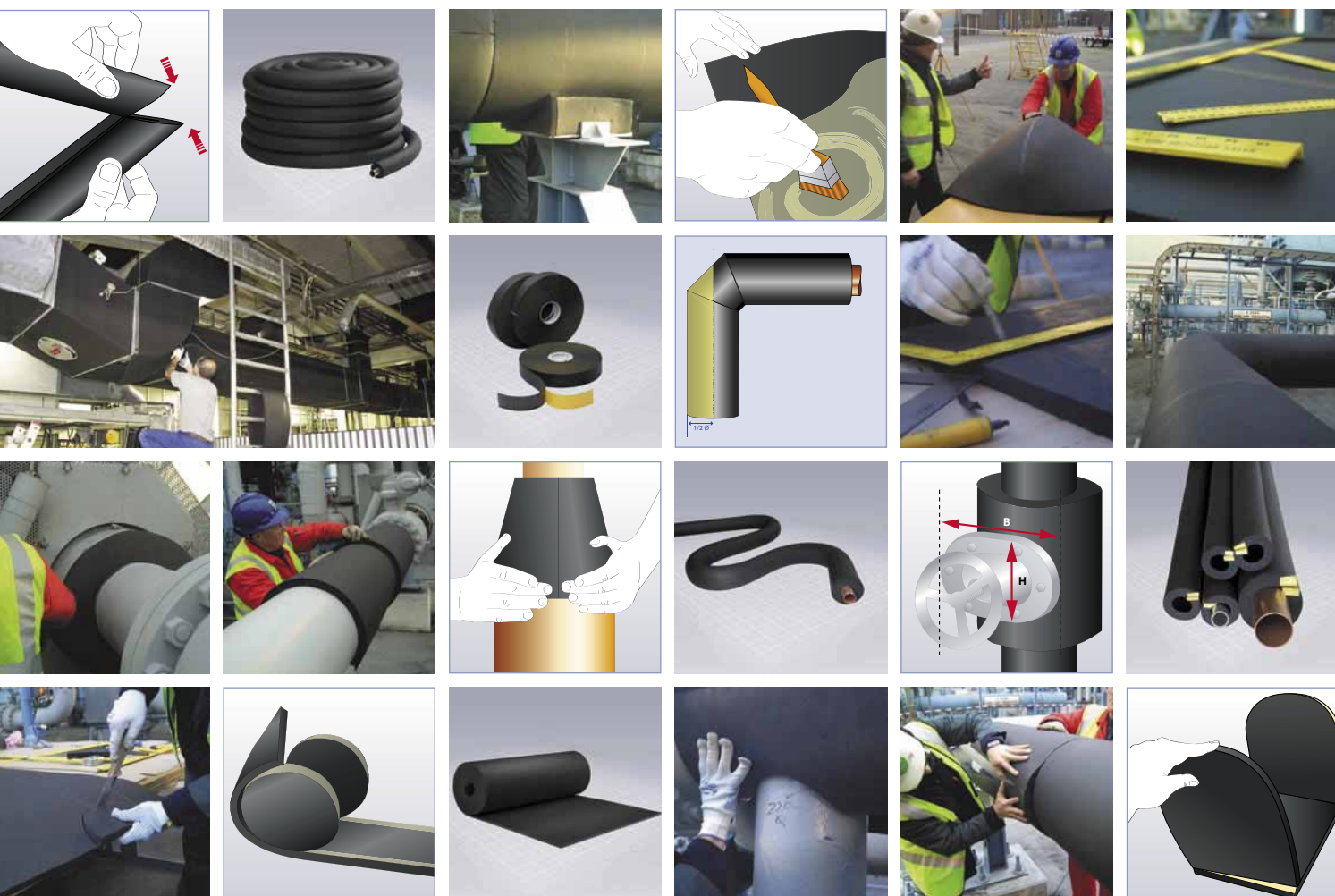




application manual

Armaflex®

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WORKING WITH ARMAFLEX

- Use good quality tools, in particular a sharp knife, fresh Armaflex adhesive and a good brush.
- Oval tubes should always be split on the flat side.
- Use clean Armaflex material – with no dust, dirt, oil or water on the surface, if present clean with Armaflex cleaner.
- Use the right dimensions! Never pull glued joints when sealing them, always push them together.
- Never insulate plants and systems that are in operation! Only start insulated plants after 36 hours - after this time the adhesive is fully cured.
- **Armafinish FR paint** can be applied immediately after the insulation has been installed, with a second coat of paint applied within 3 days, to provide UV protection (see page 5).

TOOLS FOR INSTALLING ARMAFLEX

* A three knife set plus sharpening stone are available together as a tool kit

THE CORRECT USE OF ARMAFLEX ADHESIVE**Armaflex Adhesive 520**

Armaflex Adhesive 520 has been specially developed to bond Armaflex. It joins the surfaces reliably and safely at medium temperatures of up to +105°C. The bond is resistant to weathering and aging.

Armaflex Adhesive HT625

Armaflex Adhesive HT625 has been specially developed to bond HT Armaflex insulation for medium temperatures of up to +150°C*. When using HT Armaflex only Armaflex Adhesive HT625 should be used.

* For temperatures below -50°C or above +150°C, please consult our Customer Services Department.

PREPARING FOR WORK

Check condition of Armaflex Adhesive. Cans of Armaflex Adhesive should have been stored in a cool environment wherever possible. Cans must also have been kept free from frost.

Damage due to frost can be reversed by storing in warm conditions, or for immediate use by placing the can into a bucket of hot water. Shelf life approx. 1 year.

1. Where installation surfaces are soiled with dust, dirt, oil or water all of these contaminants must be removed and, where applicable, cleaned with Armaflex cleaner before starting work. In addition all surfaces to be joined must be dry before gluing begins.
2. Pay close attention to the installation instructions on the adhesive can. Use small cans during work so that the adhesive does not thicken too quickly. Refill from larger cans when necessary and keep closed when not in use to avoid thickening.
3. Plants must not be in operation during the installation process!
4. Do not use adhesive under 0°C. If the adhesive is too cold it can be warmed in a bucket of hot water. At temperatures below 5°C, condensation can appear on the surfaces to be glued or the adhesive film. If this occurs the materials can be glued only with difficulty.
Check whether this has happened by applying absorbent paper. When working in areas with a high atmospheric humidity and high temperatures see "Hot Climates" page 33 of this manual.
5. Stir adhesive well after opening. If left to stand, heavier components in the adhesive may settle in the bottom of the can. These must be mixed thoroughly before use in order to effectively activate the adhesive.

STEEL PIPES

Check that the adhesive will adhere to any rust-inhibiting primer that has been used to protect steel pipes. Armaflex adhesive may not adhere to asphalt, bitumen, red-lead or cement surfaces.

APPLICATION

1. Use a brush with short, stiff bristles and keep clean. For larger areas a spatula or (non foam type) paint roller or the Armaflex Gluemaster may be used to speed up application.
2. Apply Armaflex adhesive thinly and evenly onto both surfaces to be glued.
3. When adhering Armaflex to other materials (e.g. metal), first apply the adhesive to the Armaflex and then to the other clean surface.
4. The tack time for Armaflex adhesives ranges between 3 and 10 min.⁽¹⁾ The time will vary according to the grade, ambient temperature and relative humidity. Avoid 'open times' ⁽²⁾ in excess of 20 min.

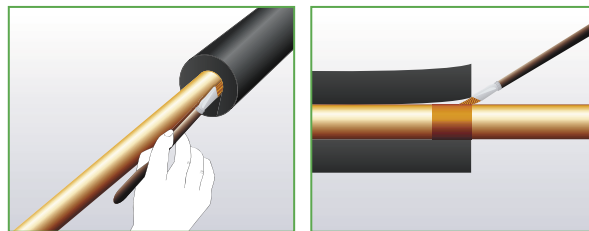
⁽¹⁾ Tack time for NH/Armaflex varies - please see special application advice for NH/Armaflex - available from www.armacell.com/uk.

⁽²⁾ The 'open time' is defined as the period between ~ first applying adhesive and finally closing together the joint, seams or surfaces.

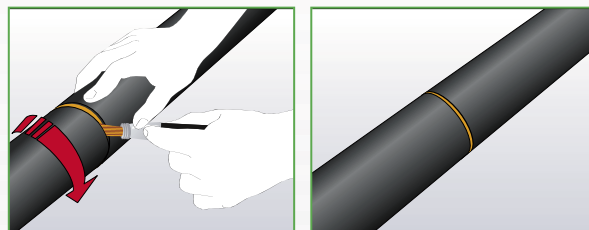
5. Allow the adhesive to 'tack-dry'. The correct initial drying time may be determined by the 'fingernail-test': touch the surface with a fingernail, if the fingernail does not adhere to the surface and the surface itself does not feel tacky the joint may be closed. The maximum adhesive force will be obtained when two tack dry surfaces are brought together.
6. The glued surfaces should be pressed together, do not stretch. Do not leave glued seams on the top of the insulation in external locations. When working outdoors, always turn the glued seams away from the sun.
7. When gluing joints under compression, with no gaps present, the wet adhesive method should be applied. Pull the seam apart slightly and apply Armaflex adhesive thinly and evenly with the brush to both surfaces and press together. No open time is needed in this case.
8. Use Armaflex cleaner to clean your tools, contaminated metal surfaces and surfaces which have had talc applied.
9. Curing time for Adhesive 520 / 625: 36 hours.

Note: Do not mix Armaflex cleaner with Armaflex adhesive to thin it out - warm it.

WET SEALING OF BUTT JOINTS



1. On all cold lines and all piping equipment in a external location, fix and secure down to the piping surface, Armaflex tube/sheet ends with Armaflex adhesive.
2. The adhesive bonding will equal the insulation thickness as a minimum.

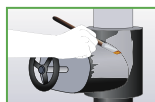


3. For the final wet sealing of the tube/sheet, pull the compressed butt joint apart with the finger and apply a thin even film of adhesive to the two butt joint edges with a small brush.
4. Apply firm and even pressure to the glued joint using the fingers and thumbs to finish.

Note: In addition all other types of Hot piping lines located externally, it is highly recommended to follow the same procedures as with cold lines.

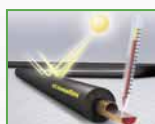
OUTDOOR USE OF AMRAFLEX

Whenever used externally Armaflex (with the exception of HT Armaflex) must be either painted, covered or clad.



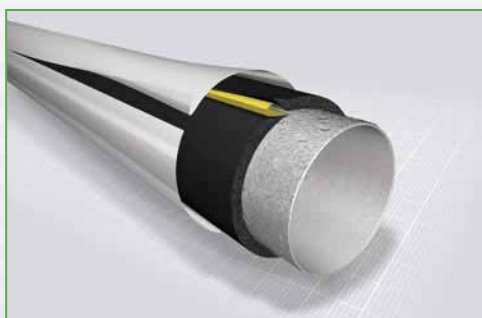
Armafinish FR paint will provide UV protection and can be applied immediately after the insulation has been installed, with a second coat of paint applied within 3 days.

	Normal consumption			
	l / m ²	m ² / l	wet film mm	dry film mm
1st coat	0.275	3.6	0.275	0.13
2nd coat	0.275	3.6	0.275	0.13
Total	0.550	1.8	0.550	0.26



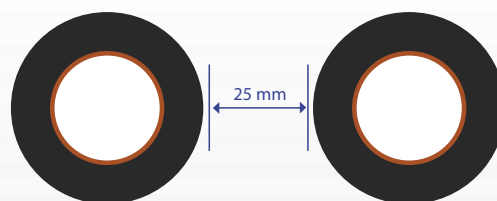
The UV-resistant **HT/Armaflex** can be used for outdoor applications without any additional UV-protection

If additional mechanical protection or protection against severe weather conditions is required **Arma-Chek covering systems** offer a non-metallic cladding option. For installation details please see the Arma-Chek installation manual.



ADVICE FOR INSULATING REFRIGERATION AND A/C EQUIPMENT

- Attach each end of the Armaflex tube to the pipe with Armaflex adhesive, and make sure the adhesive joints are firm at critical points such as flanges, T-sections, elbows, supports, etc.
- All connected items of equipment shall be insulated with equal thickness where practical.
- Never insulate chilled water lines or refrigeration equipment if the sections to be insulated are too close together. An air gap of at least 25mm should be present between the chilled water line insulated surfaces to prevent sweating.



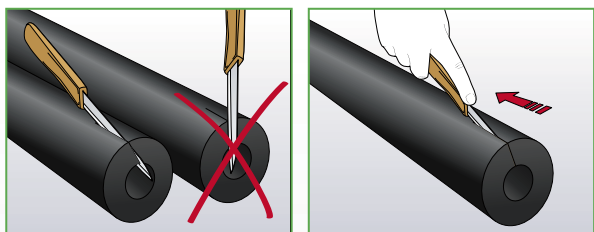
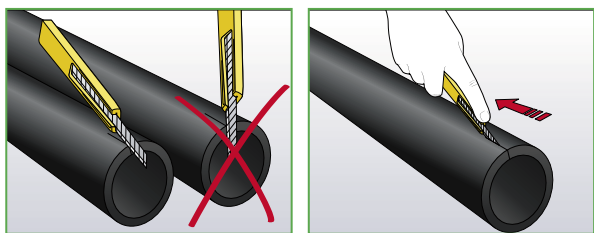
- Provide effective protection against corrosion on surface of steel pipes and vessels. Armaflex should not contribute to corrosion however, due to the presence of chloride ions in the normal building environment, it remains best practice to pre-coat the steel surface to be insulated using either aluminium foil or special paints before applying the Armaflex.

INSULATING STAINLESS STEEL PIPES

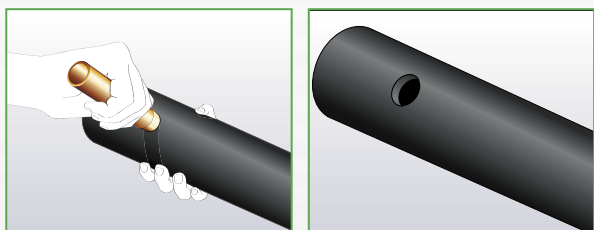
BS 5970 gives a number of specific recommendations when insulating stainless steel pipes. These installation practices greatly reduce the risk of stress corrosion cracking. All Armacell recommendations for installation on stainless steel pipework are intended to be in compliance with the procedure laid out in BS 5970.

For further information relating to the insulation of stainless steel pipes using Armaflex please see the Keytec detailed in the other references section of this application manual.

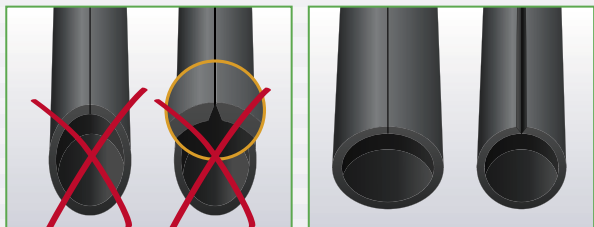
CUTTING ARMAFLEX TUBES



Use a sharp knife. Keep knife at a low angle when slitting tube



Use sharpened off-cuts of pipes to make holes.



Always cut on the flat sides of tubes.

INSULATING NEW PIPEWORK WITH SLEEVE-ON-TECHNIQUE

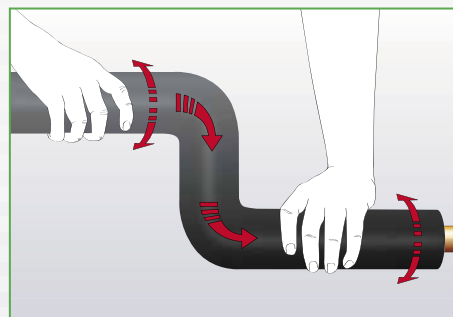
In principle, tube material can simply be slid round bends.

However, with tight bends (such as those likely to be encountered on small bore pipes) there is a risk that the insulation will kink in the throat of the bend, reducing insulation thickness.

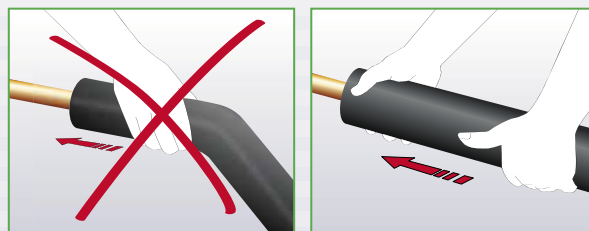
In the refrigeration/air-conditioning sector the calculated insulation thickness is then no longer achieved and condensation can occur on the surface of the insulation. When installing tubes with a self-adhesive seal there is the additional risk of compression of the adhesive lining in the bend area, which can lead to seams coming apart.

The following should be taken into account in these cases:

If the insulation kinks and the adhesive seam is compressed the bends should be cut into segments to fit (see page 11).

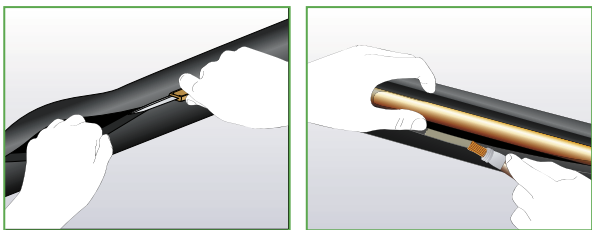


Note: Do not attempt to pull the Armaflex tube along the pipe as this may cause the insulation to tear.

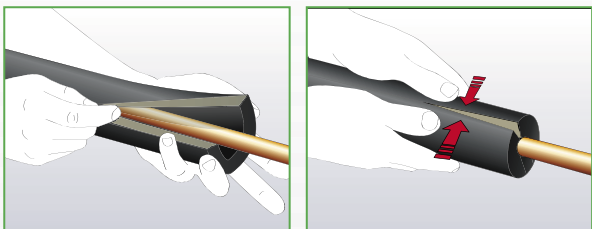


Always push the Armaflex tube over the pipe as shown.

INSULATING EXISTING PIPEWORK WITH SNAP-ON-TECHNIQUE

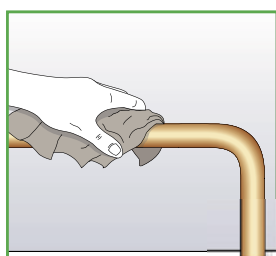
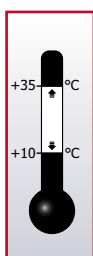


1. With a sharp knife, slit the flat part of the unslit tube along the entire length.
2. Place the slit tube onto the clean pipe; apply Armaflex adhesive to the two cut edges with a thin even film of adhesive using a short bristle brush. Apply the adhesive at 200mm intervals, along the tube length.

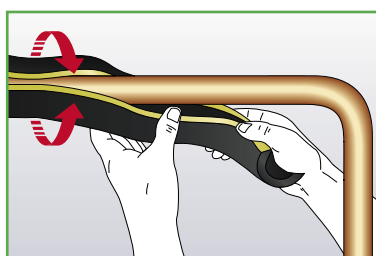


3. Allow the adhesive to touch dry, test with the fingernail.
4. Free the seams from the pipe where applicable, line the edges together and press the seam detail with firm even pressure to finish.

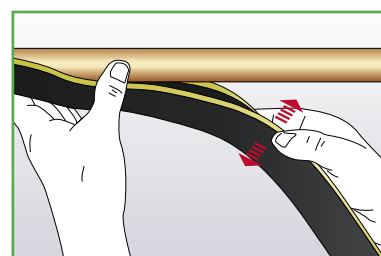
INSULATING PIPES WITH ARMAFLEX SELFSEAL TUBES



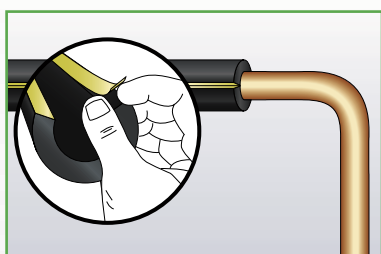
Clean all dust, dirt, oil and water from pipework using Armaflex cleaner where necessary. Install Armaflex when ambient temperature is between +10 and +35°C.



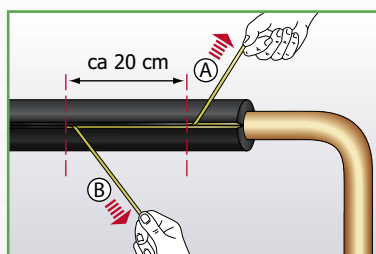
Open pre-slit Armaflex and snap onto pipe (the release paper is still protecting the self-adhesive strip)



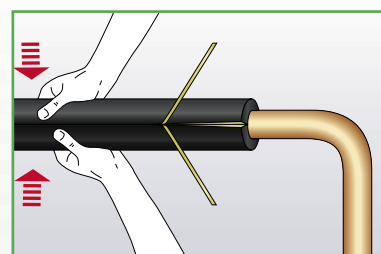
Adjust fitted Armaflex to ensure slit is easily accessible.



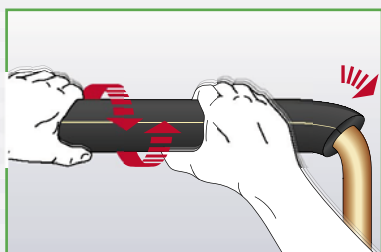
Locate the end of yellow release paper for the self-seal strip.



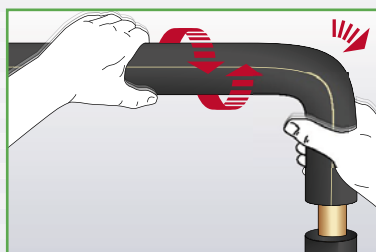
Remove the yellow protection paper strip on both sides, by drawing it away from the insulation. Please note: remove protective release paper strip from both sides!



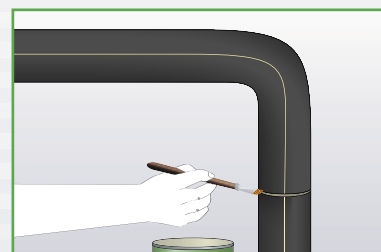
Close the slit seam and squeeze together with firm pressure, to ensure a permanent seal.



Push the Armaflex along the pipe using a circular motion. **Do not pull the insulation.**



Using a circular motion, push the Armaflex around bends. **Do not pull the insulation.**



Seal all joints with Armaflex Adhesive. **Do not stretch when sealing joints.**

Note: In principle, tube material can simply be slid over bends. However, with tight bends in pipes (small radius) there is a risk that the insulation will kink in the throat of the bend, reducing insulation thickness.

In the refrigeration/air-conditioning sector the calculated insulation thickness is then no longer achieved and condensation can occur on the surface of the insulation.

When installing Armaflex SelfSeal tubes there is the additional risk of compression of the adhesive lining in the bend area, which can lead to seams coming apart.

The following should be taken into account in these cases: If the insulation kinks and the adhesive seam is compressed the bends should be cut into segments to fit (see page 11).

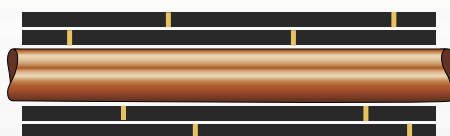
MULTI-LAYER INSULATION OF PIPEWORK

Staggering on pipe-work: cross section view



The inside diameter of the second oversize tube (where applicable) should be selected according to the maximum outside diameter of the first layer.

Staggering on pipe-work: longitudinal view



Yellow lines indicate glued seams

Important: Stagger all joints and seams

If the outer diameter of the first insulating layer is above 114mm, we recommend that the second layer be made with sheet, since this can be adapted exactly to the outside diameter of the first layer.

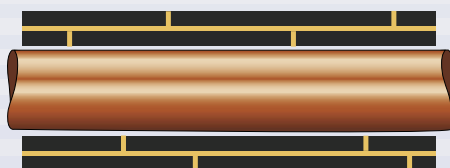
The ends of the tube or sheet of the second layer should be glued to the under layer of Armaflex.

Note: For pipework >500 mm O.D. please see page 15 "Armaflex Adhesive and its use on large bore pipework" of this manual.

Multi-layer insulation for pipework > 500 mm O.D.

- In the case of double-layer insulation the first and second layers should be glued over the entire surface. DO NOT apply only spots or strips of adhesive.
- The butt and longitudinal seams of the second layer must always be positioned offset to the first layer. The Armaflex surfaces of the first layer should be clean, dry and free from contamination. Armaflex cleaner can be used when oil or grease are present

Caution: Ensure that the underlying glued seams have cured for 36 hours, before using the Armaflex cleaner.



Yellow lines indicate glued seams

Multi-layer insulation for flat surface

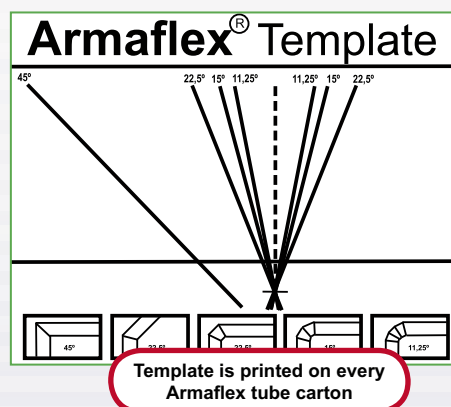
- In the case of single and multi-layering applications adhesive is applied with all-over adhesive coverage to all contacting surfaces. DO NOT apply only spots or strips of adhesive.

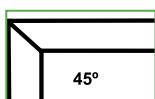
USING THE ARMAFLEX TEMPLATE

The fabrication of bends and tees using Armaflex tube requires tubes to be cut to different angles. In order to make this process easier and quicker, the Armaflex template is provided on every box of Armaflex.

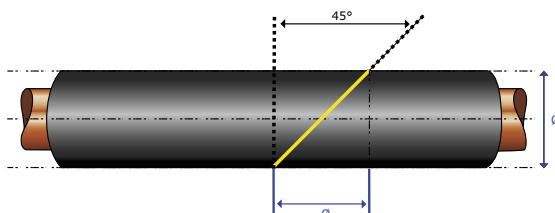
1. Place a copy of the Armaflex template face up on a table or worktop.
2. Line a tube of Armaflex across the template parallel along the horizontal base line.
3. Select the required angle cut from the template and cut along this line.

Additional copies of the Armaflex template on hard PVC sheet are available on request (contact your local Armacell representative for more information).

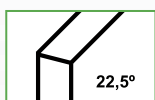




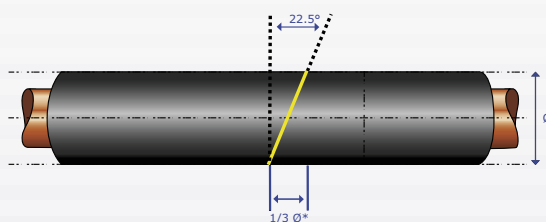
BEND WITH 90° ANGLE USING ARMAFLEX TUBE



Note: Yellow lines indicate where cuts are to be made. For correct angle measurements please use the Armaflex template located on each tube box.

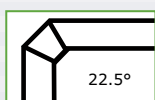


BEND WITH 45° ANGLE USING ARMAFLEX TUBE

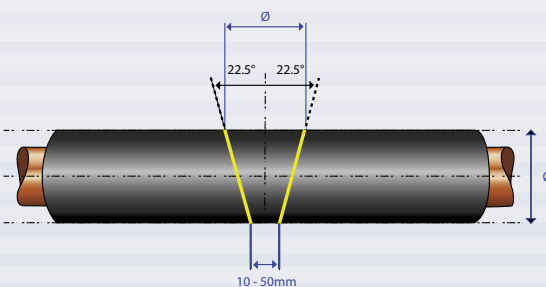


* The Ø details to achieve the 45° angle are approximate values!

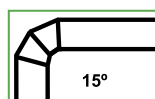
Note: Yellow lines indicate where cuts are to be made. For correct angle measurements please use the Armaflex template located on each tube box.



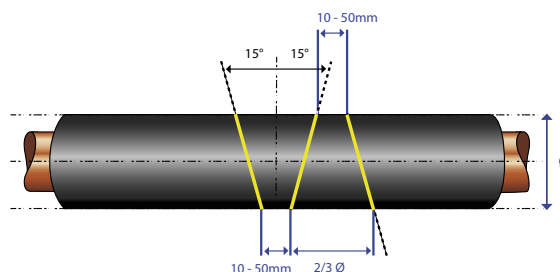
SEGMENT BEND WITH 1 MIDDLE PART - 2+1 USING ARMAFLEX TUBE



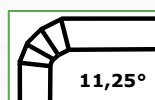
Note: Yellow lines indicate where cuts are to be made. For correct angle measurements please use the Armaflex template located on each tube box.



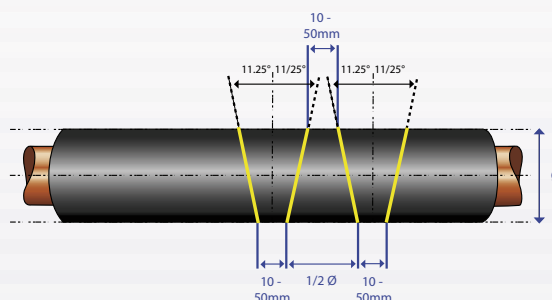
SEGMENT BEND WITH 2 MIDDLE PARTS - 2+2 USING ARMAFLEX TUBE



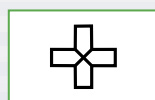
Note: Yellow lines indicate where cuts are to be made. For correct angle measurements please use the Armaflex template located on each tube box.



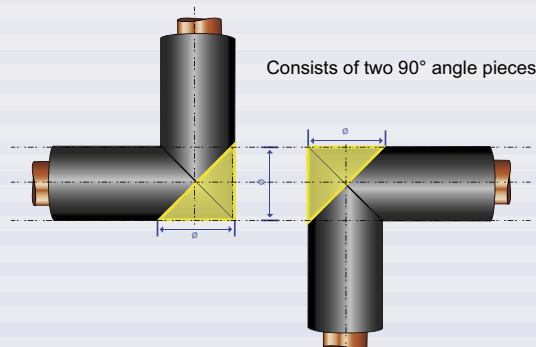
SEGMENT BEND WITH 3 MIDDLE PARTS - 2+3 USING ARMAFLEX TUBE



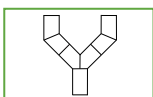
Note: Yellow lines indicate where cuts are to be made. For correct angle measurements please use the Armaflex template located on each tube box.



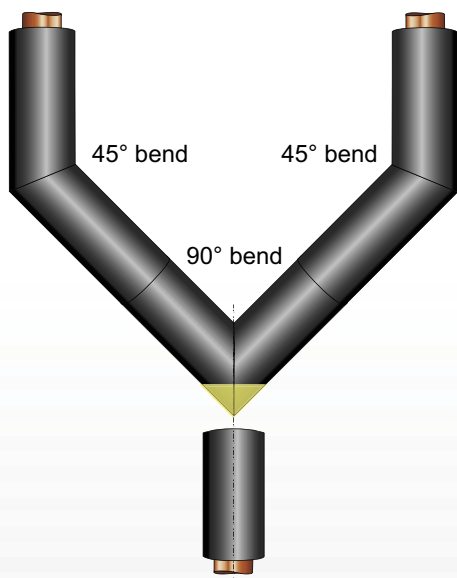
CROSSPIECE JOINT USING ARMAFLEX TUBE



Note: Yellow lines indicate where cuts are to be made. For correct angle measurements please use the Armaflex template located on each tube box.

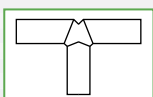


Y-TUBE USING ARMAFLEX TUBE

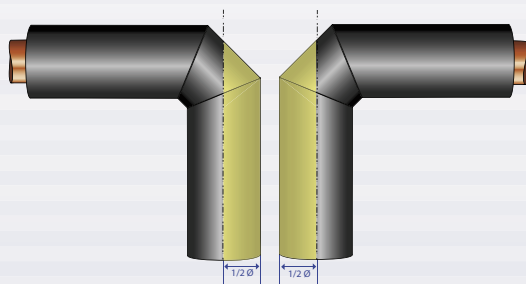


Note: Yellow lines indicate where cuts are to be made. For correct angle measurements please use the Armaflex template located on each tube box.

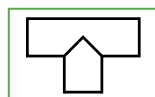
Further fabrication of the 45° bend (2 times) and the 90° bend (once).



SWEPT T-PIECE USING ARMAFLEX TUBE

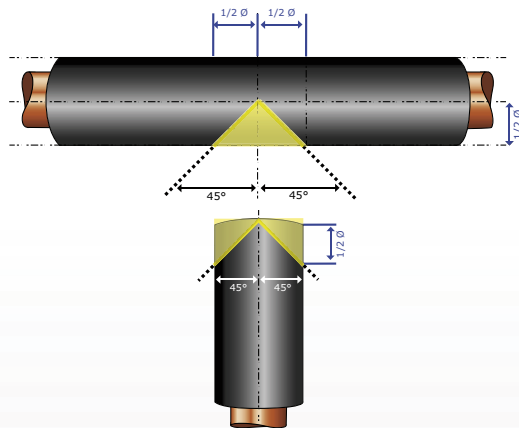


Note: Yellow lines indicate where cuts are to be made. For correct angle measurements please use the Armaflex template located on each tube box.



T-PIECE USING ARMAFLEX TUBE

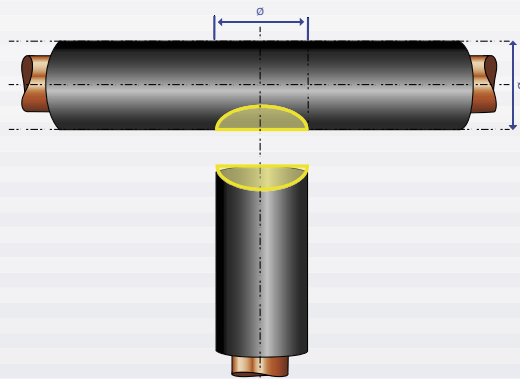
Method 1 - The "Mitre Block" T Piece



Note: Yellow lines indicate where cuts are to be made. For correct angle measurements please use the Armaflex template located on each tube box.

1. Join pre-cuts parts with adhesive to form a "T"
2. Slit open the "T", it can then be slid over the pipes

Method 2 - The "Punched Hole" T Piece

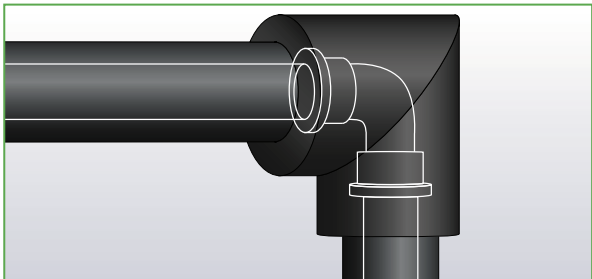


Note: Yellow lines indicate where cuts are to be made.

1. Punch a hole in the tube - with a sharpened section of a copper pipe of the right diameter - forming the crossbar of the "T"

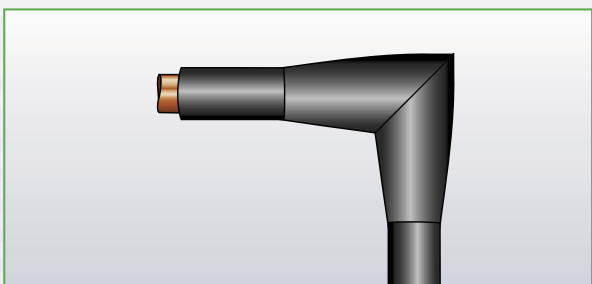
Note: For larger hole cut-outs use a pair of dividers to "mark out" and cut using a small sharp knife.

2. Slit this section of the tube open (half through the hole) and slid it over the pipe
3. Cut a semi-circular recess in the end of the branch section of tube. It is better to have a cut which is a little to deep rather than to shallow.
4. Attach the branch section to the pipe and join the two halves of the "T". Adhere all seams with Armaflex adhesive

INSULATING COUPLING PIPE JOINTS**Method 1: Oversized 90° bend**

The fitting cover is made from tube, the bore of which is the O.D. of the incoming tube. Provide a minimum 25 mm overlap at each side. The fitting may be made up using any of the methods described on page 11. Slit in the throat, apply adhesive to seams, fit when tack dry. Wet seal both over-lapping contact surfaces.

1. Push the Armaflex tube as far as the pipe fitting.
For chilled water and refrigeration pipework secure the bore of the tube to the pipe with adhesive. The adhesive band to be the same width as the insulation thickness.
2. The fitting cover is made from tube the bore of which is the O.D. of the incoming tube. Provide a minimum 25 mm overlap at each side. The fitting may be made up using any of the methods described on pages 11. Slit in the throat, apply adhesive to seams, fit when tack dry. Wet seal both over-lapping contact surfaces

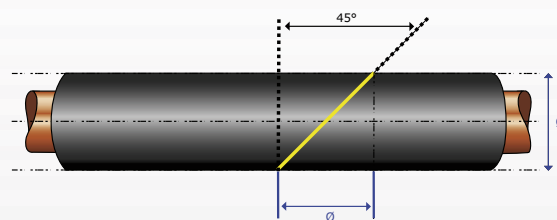
Method 2: Smooth finish 90° bend

1. The fitting cover is made from tube material, the bore of which corresponds to the maximum outside diameter of the screwed fitting. Provide at least a 38mm overlap beyond the end of the fitting on each side.
2. For pipes up to 35mm, cut two wedge-shaped pieces at 180° to each other, from top and bottom centres, at each end of the fitting. For large bore

tubes cut four wedges, one at each 90° around the circumference.

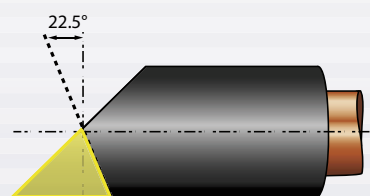
The wedges should taper back to the points where the fitting would start and be sized so that the bore of the Armaflex reduces to the O.D. of the incoming pipe. Glue the seams on the reducing sections.

3. Slit in the throat, apply adhesive, allow to tack dry, then fit. To complete, wet seal butt joints with Armaflex adhesive.

ANGLE T-PIECE (OFF-SET) USING ARMAFLEX TUBE

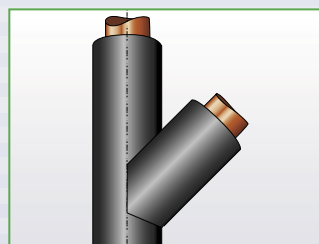
Note: Yellow lines indicate where cuts are to be made. For correct angle measurements please use the Armaflex template located on each tube box.

1. Make a 45° cut as shown above

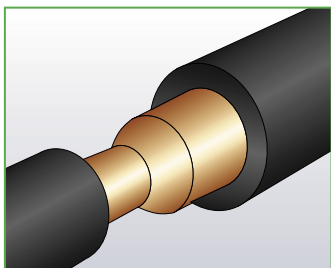


Note: Yellow lines indicate where cuts are to be made. For correct angle measurements please use the Armaflex template located on each tube box.

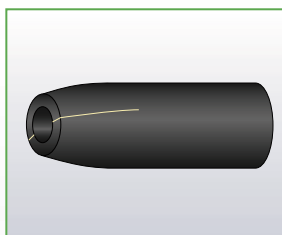
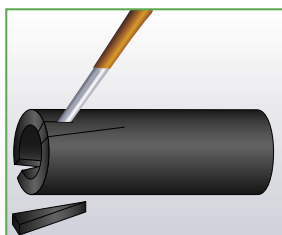
2. Use the piece of the tube with the 45° and mark a 22.5° angle and cut off as displayed above.
3. Chamfer the inside of the tube where it touches the insulation around the straight pipe
4. Wet seal all seams.



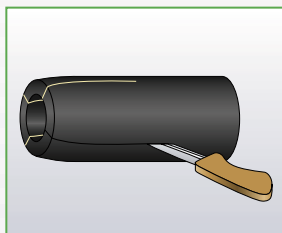
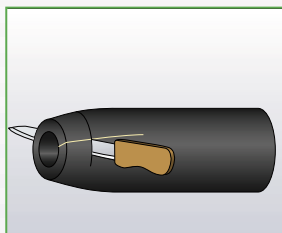
PIPE REDUCER USING ARMAFLEX TUBE



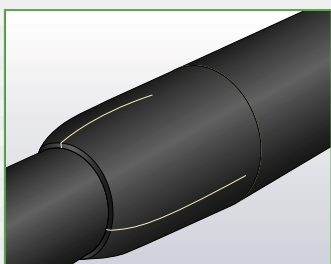
Pipe reducer to be insulated



Cut out segments and glue seams with Armaflex adhesive



Cut reducer to size - allow compression of 5mm at each end. Slit fitting on the flat side



Install and glue seam and butt joints

INSULATING PIPES WITH ARMAFLEX SHEET

Armaflex tubes are available for pipe with outer diameters up to 114mm. Larger pipes, ducts and tanks must be insulated using Armaflex sheet.

It is often advantageous to insulate smaller pipes using Armaflex sheet, even when correctly dimensioned tubes are available. Care should be taken to ensure that the stresses in the seams, caused by the bending in the sheet, do not become too great.

These stresses rise as the insulation thickness increases and as the pipe diameter decreases. Please consult the table below to gauge the applicability of different thicknesses of Armaflex sheet (recommendations may vary for HT/Armaflex and NH/Armaflex).

The ambient temperature during installation will also impact on the levels of stress likely to be encountered.

For advice on installation Armaflex sheet in ambient temperatures $\geq 5^{\circ}\text{C}$ please see table below.

Sheet Thickness	Pipe Outer Diameter / mm			
	≥ 88.9	≥ 114	≥ 139	≥ 406
9 mm	●	●	●	●
13 mm	●	●	●	●
19 mm	●	●	●	●
25 mm		●	●	●
32 mm			●	●
50 mm				●

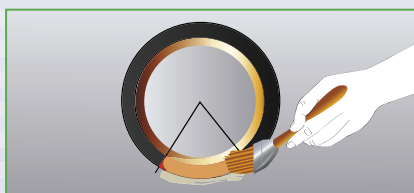
(Recommendations may vary for HT/Armaflex and NH/Armaflex. please see special application advice - available on www.armacell.com/uk)

ARMAFLEX ADHESIVE AND ITS USE ON PIPEWORK > 88.9 MM O.D.

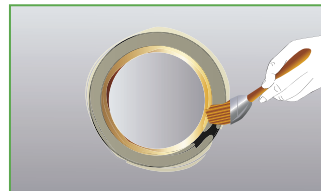
Armaflex adhesive must be applied to all seams and joints on tube and sheet fabrications up to 500 mm O.D pipe size, at each tube and sheet end Armaflex shall be adhered to the pipe.

In addition

- **when insulating horizontal pipework** with an O.D. > 500 mm adhesive should also be applied to $\frac{1}{4}$ of the pipe surface and the facing Armaflex sheet surface in order to prevent "bellowing". The picture below illustrates the area which should be covered using adhesive.

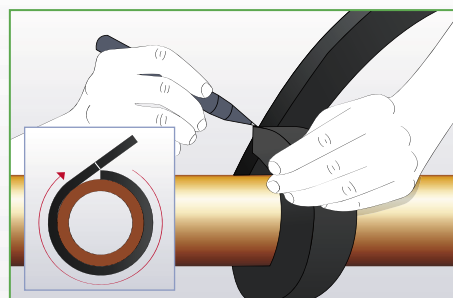


- **When insulating vertical pipework**, all-over adhesive coverage must be applied to both to the pipework and the facing Armaflex sheet regardless of pipe outer diameter.



For temperatures lower than -50°C please refer to special application advice "Insulating low temperature lines"

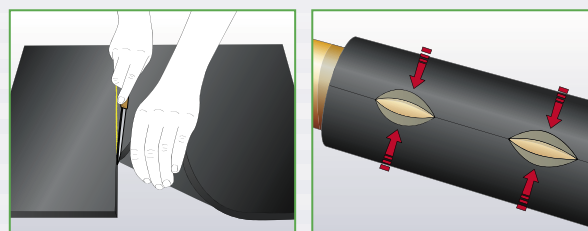
INSULATING LARGE PIPES WITH ARMAFLEX SHEET



1. Determine the circumference of the pipe.

Important: Always measure with a strip of Armaflex of the thickness to be used for the insulation.

Warning: Do not stretch the strip.



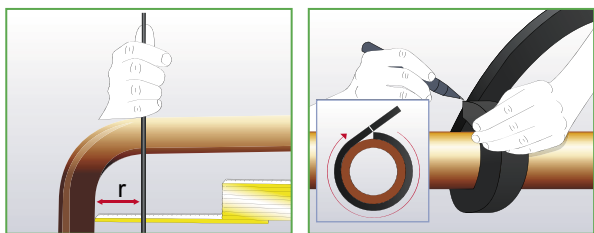
2. Cut Armaflex sheet to the required size - apply Armaflex adhesive to the cut surfaces in a thin layer, allow to touch dry.

3. Press together at the ends and then in the middle. Close the entire seam starting from the middle.

Note: In order to prevent the seam re-opening ensure the adhesive has been fully applied to the edges of the fixing seam and ensure the correct amount of adhesive has been applied.

Check the open time of adhesive to ensure it is still fit for use.

TWO-PART-BEND WITH ARMAFLEX SHEET

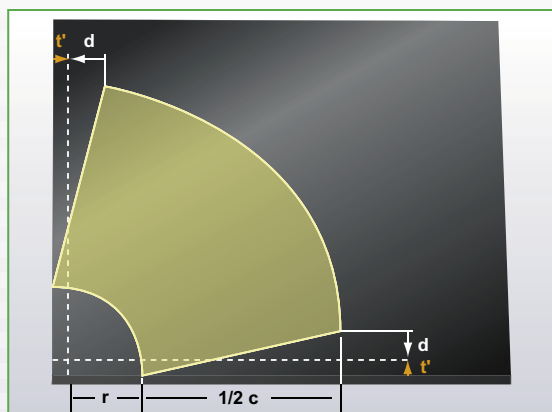


Establish the inside radius, r , by dropping a perpendicular line to meet a horizontal line from the outside of the two welds. The point where these two lines intersect gives the origin for the radius, r . This is the throat radius.

Measure in a trimming allowance equal to the thickness of insulation being used along both vertical and horizontal edges then transfer r to the sheet as indicated.

Determine the circumference of the pipe using a strip of Armaflex of the thickness to be installed.

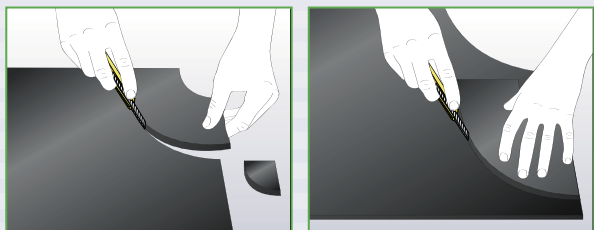
Note: Do not stretch



Halve the pipe circumference and transfer this dimension to the Armaflex sheet.

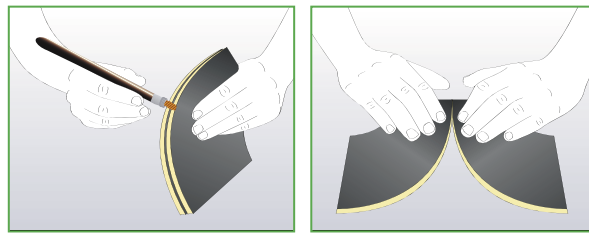
Mark out the two arcs from the intersection of the trim lines.

- r = inside radius of bend
- $\frac{1}{2} c$ = half of pipe circumference
- t = insulation thickness (in mm)
- d = $\frac{1}{4}$ pipe diameter



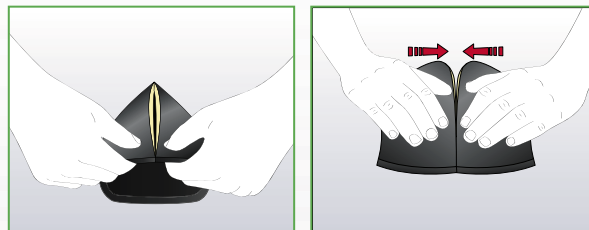
Cut out the first half-section of the elbow.

Use the first half-section as a template to cut out the second half-section of the elbow.



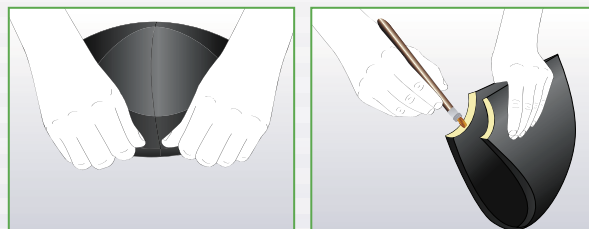
Place the sections together with the rough surfaces inwards. Apply Armaflex adhesive to the outer edges.

Allow the adhesive to tack dry (fingernail test) then press the two sections together at one side to make a short seam.



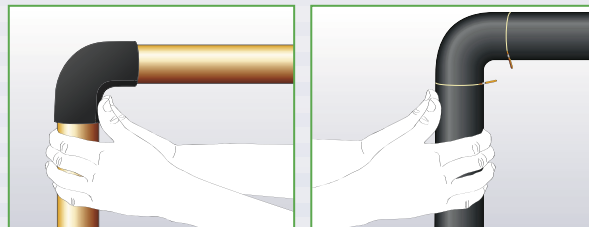
Next, press the opposite sides together, also making a short seam. Repeat alternately closing 50-75mm at a time on each side, working towards the centre.

Press the remainder of the joint firmly together.



Turn the assembly over and press the seam firmly together from the inside, so that a good adhesive joint is achieved across the entire wall thickness.

Apply Armaflex adhesive to the inner joint edges.

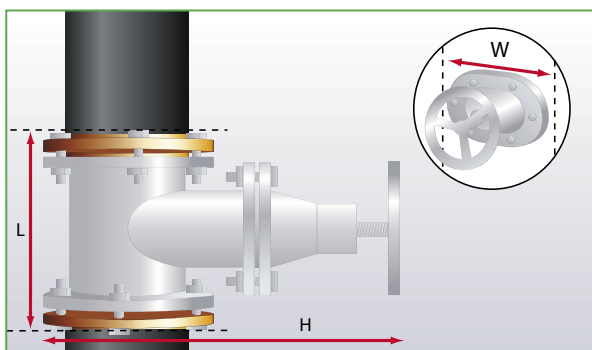


Place the insulation cover over the pipe bend. Allow the adhesive to tack dry then press the joint faces firmly together.

Wet seal jointing details with adhesive, fitted under slight compression, to complete the bend.

INSULATING VALVES WITH D-BOX MADE OF ARMAFLEX SHEET

Recommended method

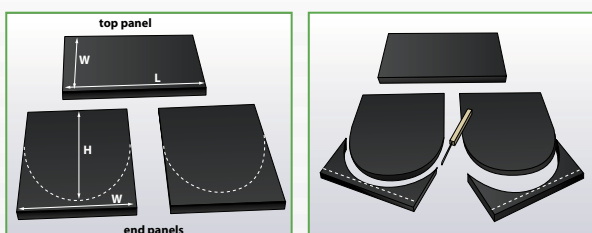


Establish the following measurements:

L = length of valve + 2 x thickness of insulation

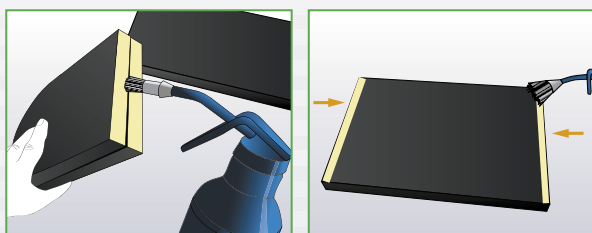
H = height of valve + 2 x thickness of insulation

$W = \varnothing$ (diameter) + 10mm



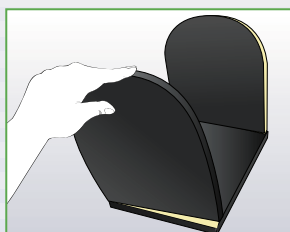
Mark out and fabricate 2x end panels and 1x top panel using the measurements made in the previous step.

Cut cleanly using a small sharp knife.



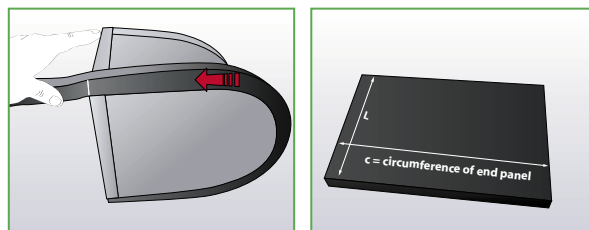
Apply Armaflex adhesive along the edges as indicated.

Note: The glue line must be as wide as the thickness of the Armaflex in use.



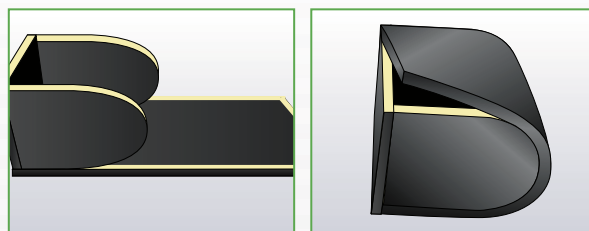
Glue the top edges of the end panels and the top panel edge.

Fix down the end panels to the top panel making sure the edge's are in-line



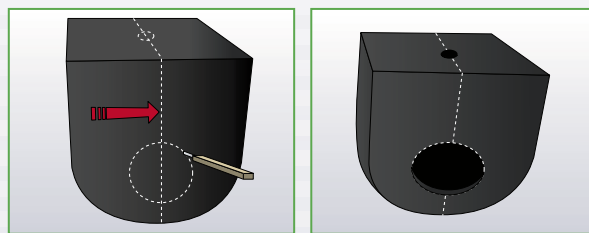
Use a strip of Armaflex (used thickness) to determine the circumference around one end panel (including the top panel).

Mark measurement L and circumference out and cut the body panel to size. Apply Armaflex Adhesive to the body panel end and the body panel edges as shown.



Gently roll the body panel edges around the end panels until the cover panel resembles a box.

Fix down the square 90° edge as shown. Ensure the edge is in-line and neat. Continue to fix all edges in this way.



Cut holes for the insulated piping connections on each of the end panels and a final cut-out for the valve spindle connection at the top.

Split the box into two halves and fit around the valve.

If required the fitting cover can be slit through three quarters of the way round the fabrication cover and snapped over the fitting - with adhesive applied to all fixing seam details as stated.

To finish, apply Armaflex adhesive to the fixing seams, allow to touch dry and fix the seams together.

Vapour seal the connections (joints) to the linear insulated pipes using Armaflex adhesive.

Note: For external installations, weather-seal around the valve spindle connection cut using Arma-Chek black mastic.

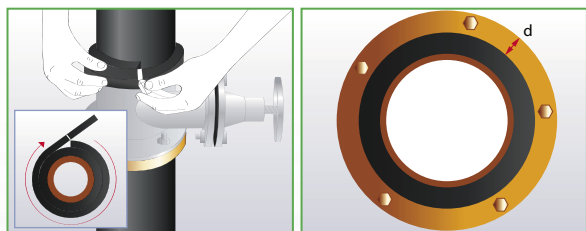
VALVE INSULATION WITH ARMAFLEX SHEET

Traditional method

Small bore heating and plumbing valves or stopcocks may be insulated using oversized tube sections.

The following section gives the general techniques for insulating larger valves and gate valves.

It is recommended that the valve body is packed with strips of Armaflex insulation stuck onto the valve body with Armaflex adhesive.

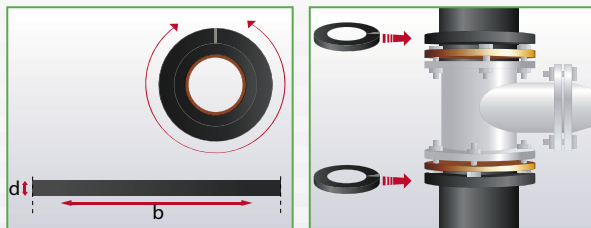


Insulate the pipe as far as the flange. Where applicable, allow for bolt removal.

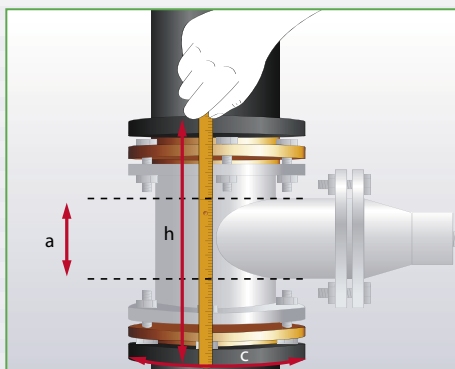
Determine the

b = circumference of insulated pipe

d = depth of flange ring



Mark out and cut a strip. Adhere both ends and apply around the flange ring (skin side of Armaflex should face top).



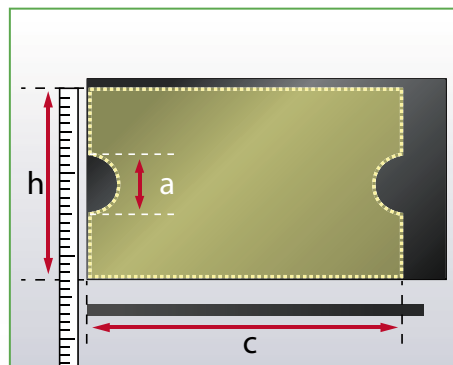
Measure

h = height between the outer faces of the two rings.

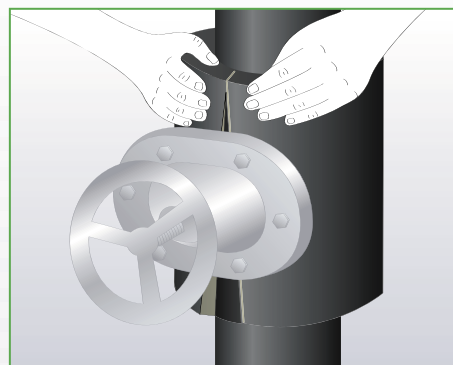
a = diameter of spindle neck

c = circumference of the rings

Important: Always measure with a strip of Armaflex of the thickness to be used for the insulation. Do not stretch the strip.



Transfer height (h), circumference (c) and diameter of spindle neck (a) to the Armaflex sheet and mark on the cut outs for the spindle neck.

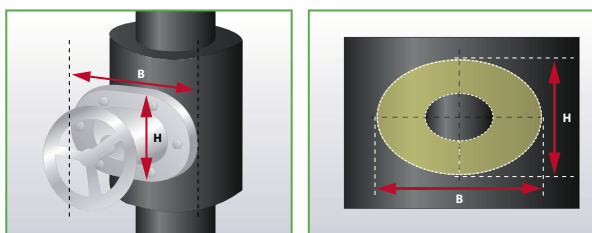


Apply a thin coat of Armaflex adhesive to all fixing seams of the valve cover.

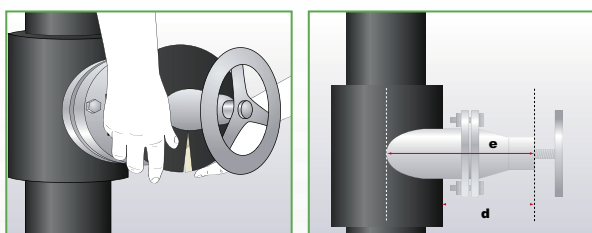
Allow to tack dry (fingernail test), fit, then press together firmly.

Note: The valve body should always be insulated after the insulated pipe has been completed.

NECK-T / PIPE-T / SPINDLE NECK OF VALVE INSULATION WITH ARMAFLEX SHEET



Measure the height of the spindle housing flange and its width, and make up an end disc.



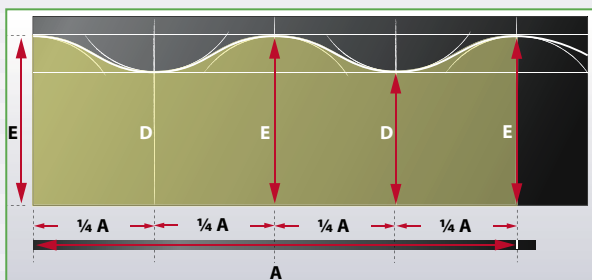
Cut the disc through at one side, apply adhesive to the cut edges, allow to tack dry.

Join the end disc after placing it over the spindle neck.

Measure

- A = circumference of the end disc
- d = minimum depths of spindle housing
- e = minimum depths of spindle housing

Important: Always measure with a strip of Armaflex of the thickness to be used for the insulation. Do not stretch the strip.

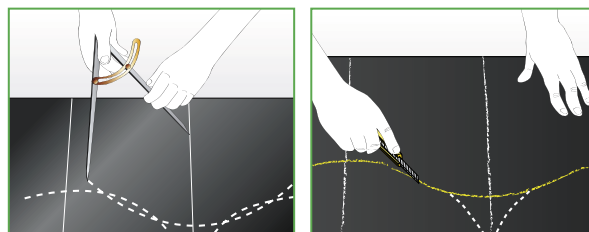


Mark off the circumference (A) into 4 equal sections.

Transfer these heights

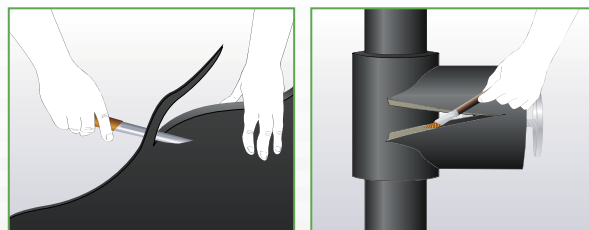
- D = d + thickness of insulation
- E = e + thickness of insulation

to the Armaflex sheet.



Using the difference between the two heights as a radius, mark out 5 arcs round the intersections of the lines, and join the arcs with a continuous line.

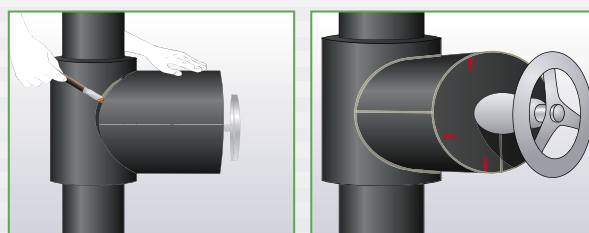
Cut out the shaped section of sheet.



Chamfer away the inside surface at the highest point (where the sheet rests against the side of the valve body).

Apply adhesive to the longitudinal seam, allow to tack dry, then seal round the spindle housing.

Note: In external locations, to provide additional protection against water entry, Arma-Chek Mastic Black can be applied around the spindle neck tee covers termination point.

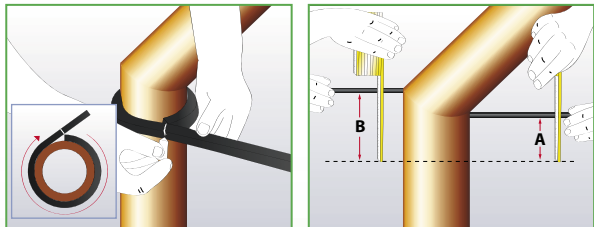


Apply adhesive using the wet seal method to all seams. The valve insulation is now complete.

Note: D-Boxes, using Armaflex sheet, can also be used to insulate valves.

OFFSET ANGLE & PIPEWORK BEND ANGLE JOINTS

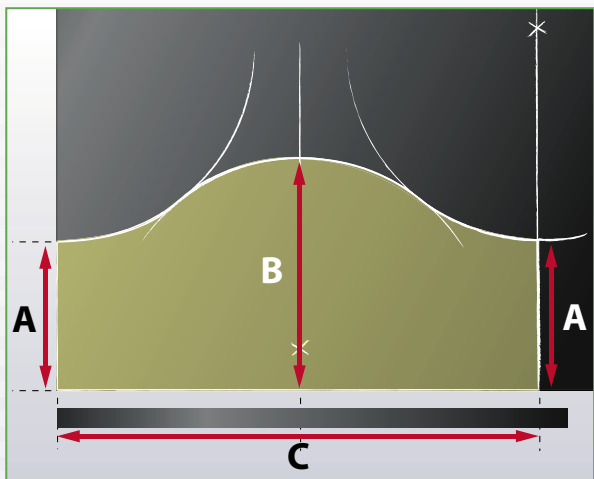
The following illustrations show the various stages of work when insulating a mitre angle or bevel joint in a pipe. The procedure when insulating a right angle pipe joint is effectively the same.



Determine the circumference of the pipe (c).

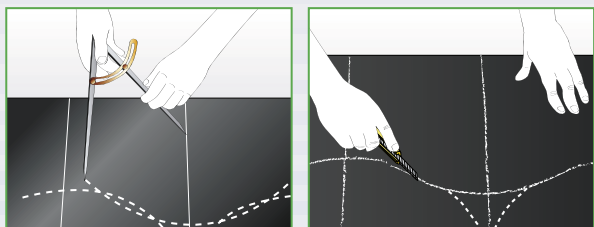
Important: Always measure with a strip of Armaflex of the thickness to be used for the insulation. Do not stretch the strip.

Measure the outer height (B) and the inner height (A) of the mitre joint.



Transfer the circumference to the Armaflex sheet and mark in the centre line.

Transfer the outer and inner height to the Armaflex sheet.

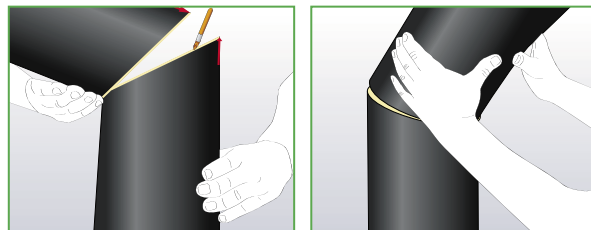


Measure the half-circumference using dividers and mark out 3 arcs.

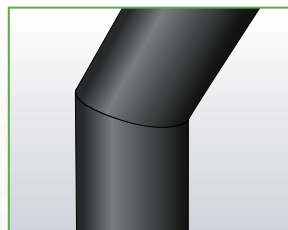
Join the arcs with a continuous line.

Cut along the line. When repositioned by 180°, the upper and lower sections produce...

...the two parts of the mitre joint.



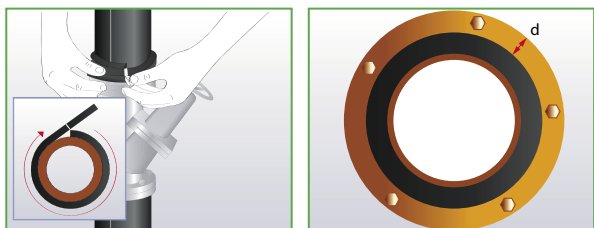
Apply Armaflex adhesive to the longitudinal seam, then to the connecting seam.



The insulation is now complete.

STRAINERS, STRAINER VALVES & INCLINED SEAT VALVES

The work involved in insulating a strainer valve or an inclined seat valve is similar (some measurements need to be extended) except that an end disc may be required.

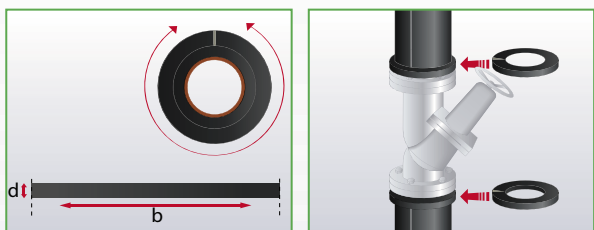


Insulate the pipe as far as the flange.

Determine the

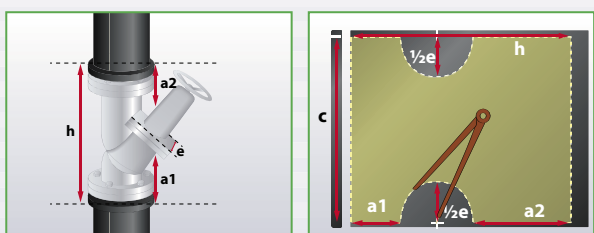
b = circumference of insulated pipe

d = depth of flange ring



Mark out and cut strip. Adhere both ends and apply around the flange ring (skin side of Armaflex should face top).

Note: It is often desirable to apply strips of Armaflex directly to the strainer at this point. Packing out in this way can add additional strength to the fitting cover and can reduce the impact of shrinkage at low temperatures.



h = Measure the distance over the outer faces of the two Armaflex rings you have positioned next to the flanges.

$a1$ = Measure the distance from the strainer to the outer face of the lower ring

$a2$ = Measure the distance between the strainer and the outer face of the upper ring

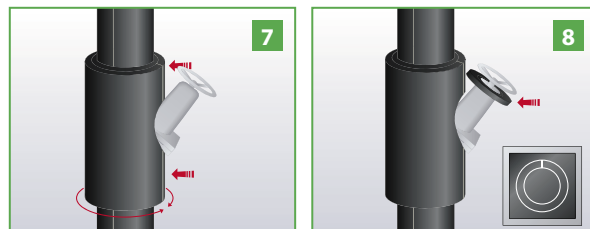
e = depth of strainer

c = Circumference of the rings.

Important: Always measure with a strip of Armaflex of the thickness to be used for the insulation.

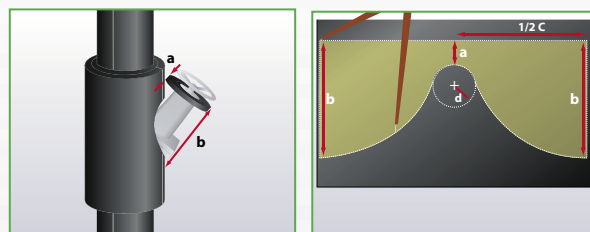
Warning: Do not stretch the strip.

Transfer these measurements to the Armaflex sheet and mark the cut-out required for the strainer.



Cut the sheet and attach the insulation to the strainer body using Armaflex Adhesive.

Cut a ring of Armaflex with inner diameter equal to the outer diameter of the insulated offset part of the strainer. Attach this ring at the end of the insulated section as shown using Armaflex Adhesive.



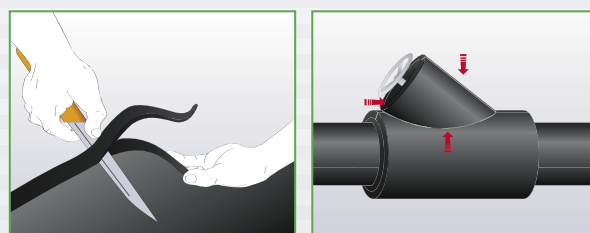
Determine the

a = shortest distance from the ring of Armaflex to the insulation around the strainer body.

b = longest distance from the ring of Armaflex to the insulation around the strainer body.

d = 1/4 diameter of the insulated valve body

Using the circumference of the offset section of the strainer prepare a sheet of Armaflex.



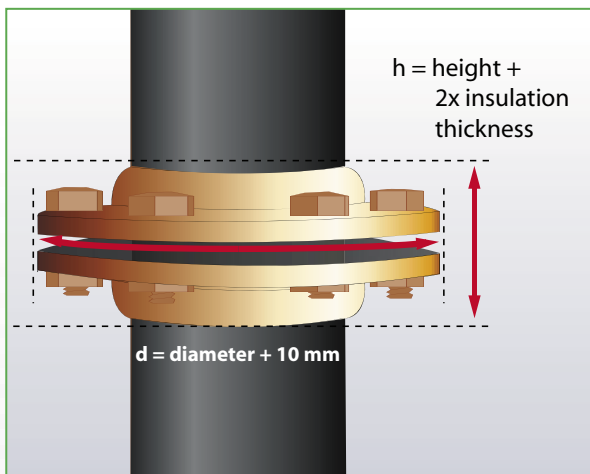
Cut the remaining section of sheet

Chamfer the sheet away where it is to touch the insulation around the strainer body.

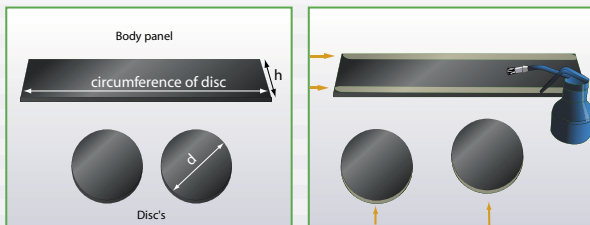
FLANGE BOXES

The following section gives the installation techniques for insulating flanges.

On chilled water or refrigeration applications it is advisable to pack the gaps between the nuts with strips of Armaflex insulation.

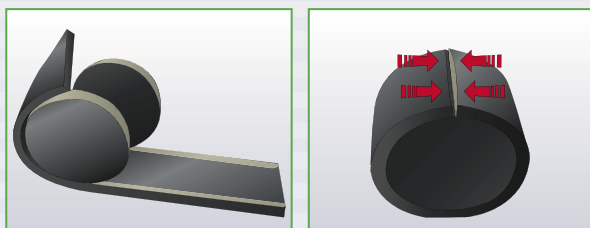


Using a pair of callipers, determine the diameter of the flange face. Please add 10mm to this measurement. Measure the length of the flange (incl. Bolts) and add 2x the insulation thickness of the used sheet. Where applicable allow for bolt removal.



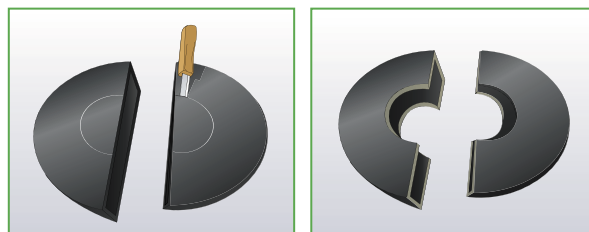
Transfer these measurements to a piece of Armaflex sheet. Mark out two concentric circles with dividers. Repeat on a second piece of sheet. Cut out two Armaflex rings.

Determine the circumference of the disc.



Roll the body panel up and around the end disc's, do not stretch during application. Check alignment throughout.

Place the edge to the a joining seam's edge opposite.



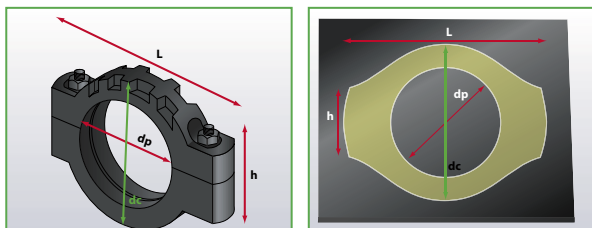
Using a small sharp knife cut out for the insulated pipe diameter.

To finish fit the two half's of the flange box around the flange and wet seal all seams and joints to the insulated pipe branch.

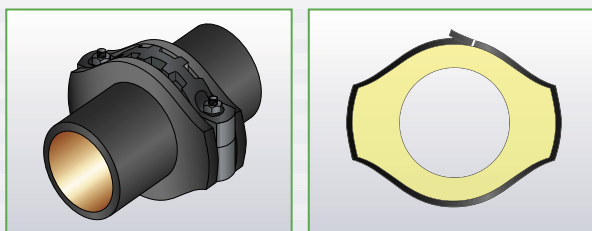
If required the fitting cover can be snapped over the fitting, by only cutting the insulation cover half way through.

VICTAULIC COUPLINGS

1. Insulate pipes up to the coupling.



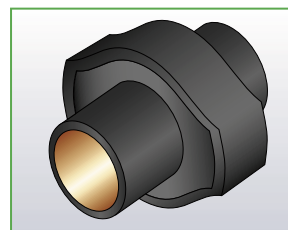
2. Determine
 d_c = diameter of coupling + 2 x insulation thickness
 h = height of screws + 2 x insulation thickness
 L = length of coupling
3. Using $\frac{1}{2}$ of d_c (diameter coupling + 2 x insulation thickness) as a radius transfer a circular arc to the Armaflex sheet and mark a horizontal centre line.
4. From the centre of the line mark the width of the coupling.
5. At both ends mark out the height of the screws plus 2x insulation thickness at a 90° angle to the centre line.
6. Connect the four endpoints and the circular arc with a tangent so that a oval like disc is built.
7. Determine the diameter of the insulated pipe and mark it on the Armaflex sheet.
8. Cut out this disc and use as a template to create a second identical disc.



9. Adhere both discs immediately next to the coupling as shown.
10. Determine the circumference of the disc and measure the distance over the outer faces of the two disc's.

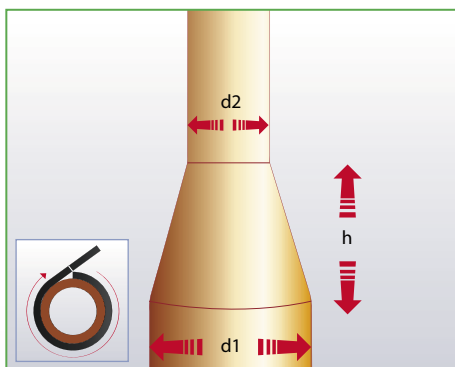
Transfer these measurements to a sheet of Armaflex.

Important: Always measure with a strip of Armaflex of the thickness to be used for the insulation. Do not stretch the strip.



11. Cut this section and adhere over the Armaflex discs around the coupling.

CONCENTRIC REDUCERS

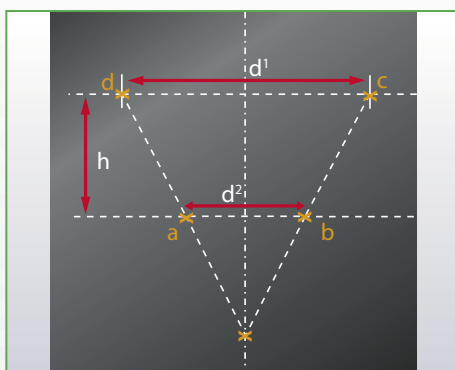


Determine the following measurements

h = height of the reducer, incl. both welds

$d1$ = diameter of larger pipe + 2 x insulation thickness

$d2$ = diameter of smaller pipe + 2 x insulation thickness

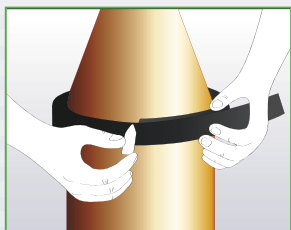
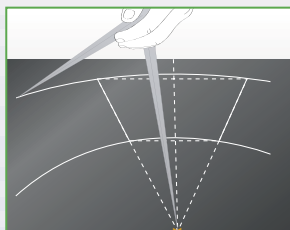


Mark out the Armaflex sheet with a centre line .

$d1$ and $d2$ are marked off at each end, to give points a,b,c and d (yellow markings show meeting points).

Distance between the lines $d1$ and $d2$ is height h .

Extend the lines d-a and c-b to meet at the apex point which is on the extension of the centre line.

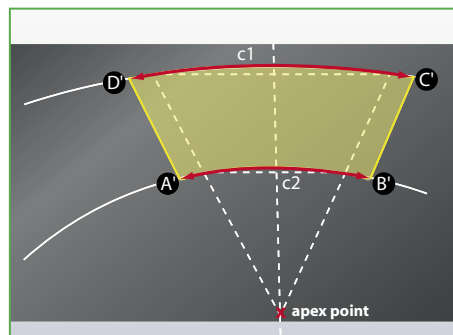


From the apex point strike two arches through a-b and d-c.

Determine the circumference of the $c1$ (large pipe) and $c2$ (small pipe) .

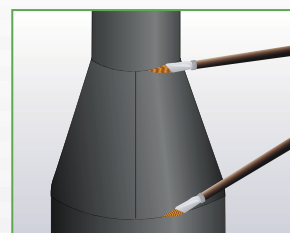
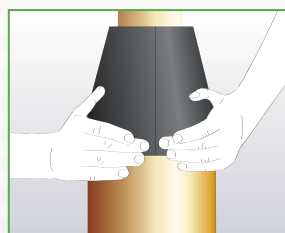
Important: Always measure with a strip of Armaflex of the thickness to be used for the insulation.

Warning: Do not stretch the strip.



Transfer the two circumferences by using the two strips used to measure the circumferences and mark the final dimension of the insulation of the reducer.

Cut out the reducer piece with a sharp knife (yellow area indicates the cutting lines)



Apply a thin coat of adhesive to the edges to be joined, allow to tack dry. Press together firmly at one end, then at the other end and complete the joint.

Complete insulation by insulating the pipes on either side of the reducer and wet seal both butt joints.

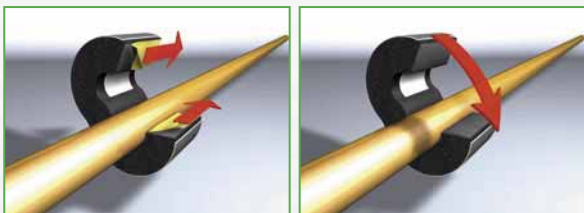
INSULATING PIPE SUPPORTS

Where Armaflex is installed, the use of Armafix or Armaload pipe supports are the preferred solution for the prevention of ice and condensation on cold installations. However, when Armafix pipe supports are not selected, the following instructions are imperative:

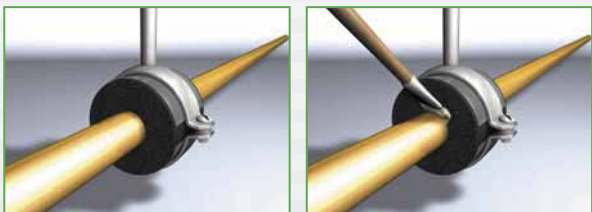
- Ensure that the Armaflex is fitted tight to the support, with no air gaps present and vapour sealed with Armaflex adhesive.
- Install Armaflex self adhesive tape to the butt joint where the Armaflex and the support meet. Ensure the surface of the butt joint is free from dust before installing the Armaflex tape.

INSTALLING ARMAFIX INSULATED PIPE SUPPORTS

Armafix pipe supports are sections of Armaflex with load bearing PUR/PIR inserts and an aluminium outer shell cladding.



Note: Select the correctly dimensioned Armafix pipe support (min. insulation thickness of tube).



Use a standard pipe support bracket around the outer aluminium shells.



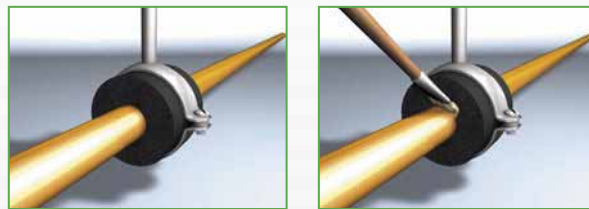
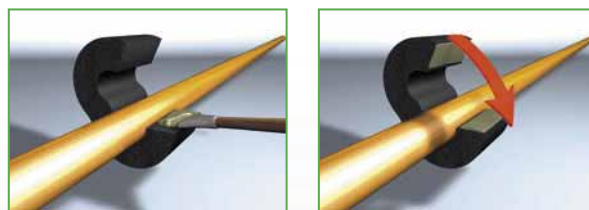
Install the Armaflex insulation on either side of the Armafix pipe support. Wet seal the butt joints with the Armafix pipe support using Armaflex adhesive.

Note: Ensure that the pipe insulation is installed under slight compression.

INSTALLING ARMALOAD PIPE SUPPORTS

Armaload pipe supports are sections of high density elastomeric insulation with **no** outer aluminium shell. Armaload pipe supports are suitable for use on heating, hot water and domestic cold water pipes.

Note: Contact Armacell customer services before installing Armaload pipe supports on chilled water and cold pipes.

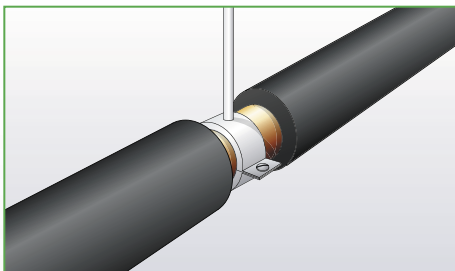


Note: Ensure that the pipe insulation is installed under slight compression.

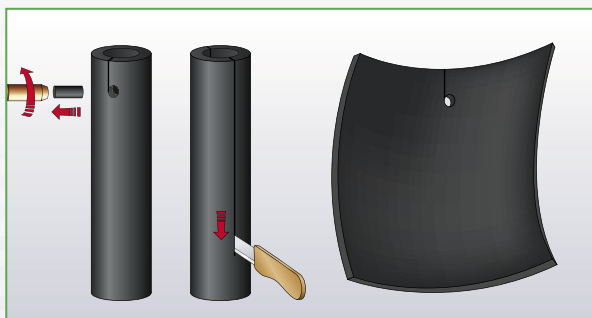
INSULATING "OVER" (ENCAPSULATING) PIPE SUPPORTS

On cold lines pipe supports must be insulated to prevent condensation. If no insulated pipe supports have been specified the following approach should be followed as a minimum precaution.

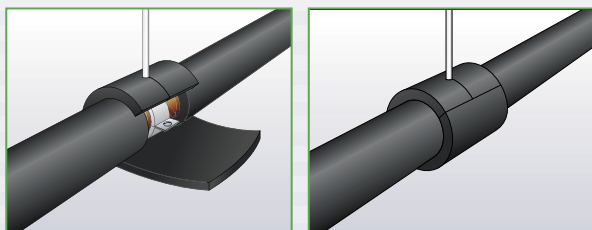
Note: It is best practice to always have an insulated pipe support such as Armafix or Armaflex installed.



1. Install the Armaflex as close to the fixing bracket as possible. Seal the ends of the tube to the pipe with Armaflex adhesive.



2. With a large off-cut of Armaflex tube, core out a small hole to allow for the oil thread support of the bracket and slit with a small sharp knife along the flat face of the tube.



3. Place the Armaflex cover over the support area, mark and cut the true circumference of the cover.

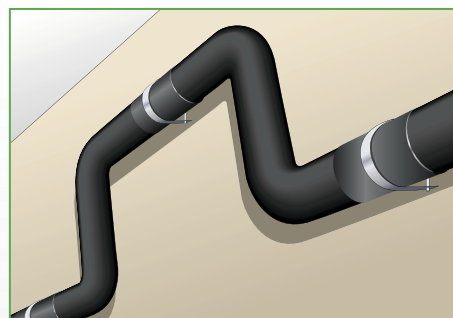
Fix and vapour seal all seams and joints in and around the attached insulation using Armaflex adhesive.

Note: Armaflex 3mm Insulating self adhesive tape can be applied 50-100mm along the attached oil thread support.

INSULATING THROUGH MOUNTED "SADDLE" SUPPORTS

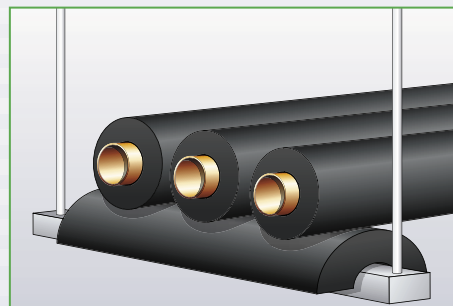
Insulate throughout continuously using Armaflex tube / sheet. Where the mounted "saddles" are located as shown, install two half (galvanized) metal sleeves around the insulated pipe.

Note: 1 - 1.5mm thick galvanized sheet will be suitable. If using metal sheet with a highly reflective, low emissivity, surface the insulation thickness may no longer be sufficient to prevent condensation.



INSTALLING ARMALOAD PIPE SUPPORT SECTIONS

Armaflex is a high density load bearing grade of Armaflex in the form of a one metre long, half round section. It is designed to be used as a load bearing insert between overhead support systems, such as unistruts, and banks of small bore copper pipes insulated with Armaflex.



Use only with copper pipes up to 54mm O.D.

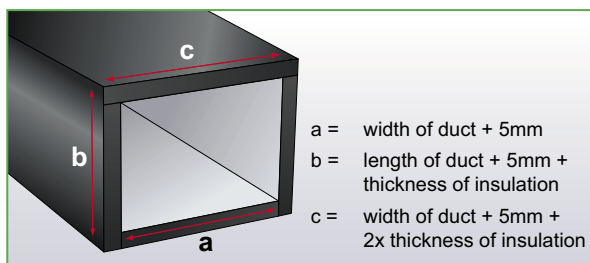
Suggested hanger spacing:

Outer pipe - Ø mm	Distance of hangers
≥ 35	3 m
42	2 m
54	1.5

Some compression will occur on both the Armaflex and the Armaflex, but this is acceptable and has been allowed for in the design principal.

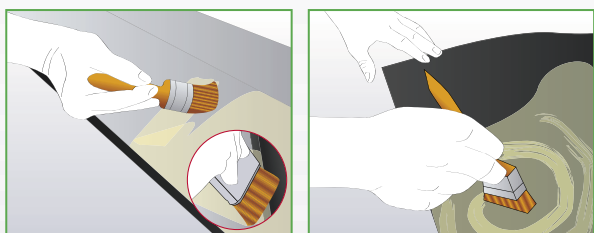
MEASURE SURFACE DIMENSIONS FOR INSULATING RECTANGULAR DUCTS

Measure surface dimensions and cut Armaflex sheet to size. **Note:** Add 5mm so that material is fitted under compression.

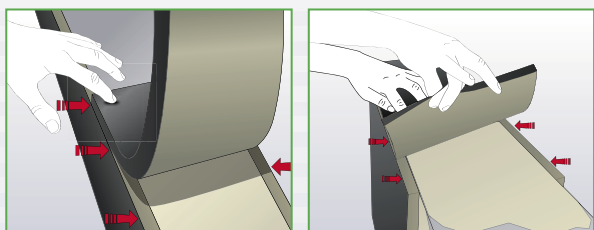


INSULATING RECTANGULAR DUCTS WITH ARMAFLEX SHEET

Clean all surfaces using Armaflex cleaner to remove grease, oil, dirt etc. and cut sheets to size

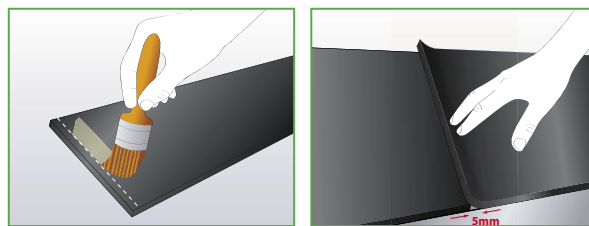
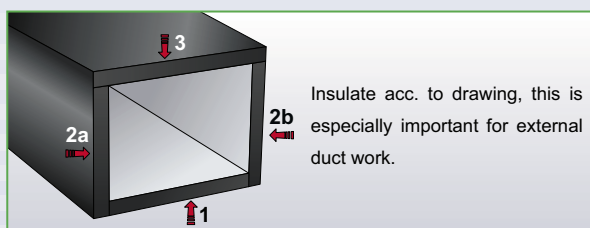


Spread a thin film of adhesive onto the metal surface and then onto the Armaflex sheet.

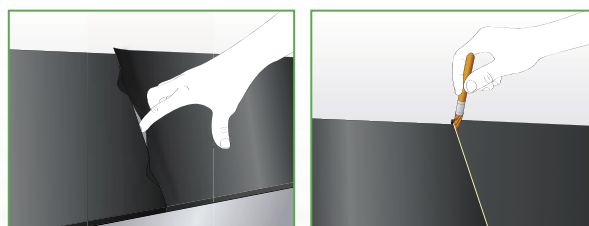


When the adhesive is tack dry (fingernail test) place Armaflex sheet in position and press firmly to achieve a good bond. Continue, applying Armaflex adhesive to both surfaces, including the Armaflex edge, and allow to tack dry before pressing firmly into position.

Note: Remember to roll the sheet down into position along the insulated edge's.



The cut sheets should be positioned so that there is a 5-10mm overlap (for compression). Do not apply adhesive to this area..

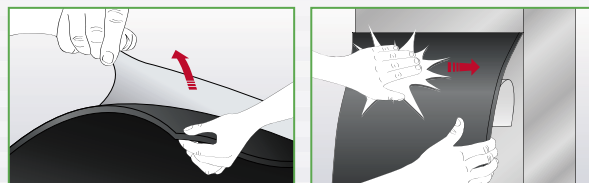


When pressed together the material is under compression and is not stretched.

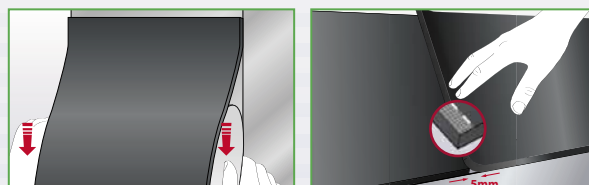
Apply an additional wet seal along the butt joints.

INSULATING RECTANGULAR DUCTS WITH ARMAFLEX SELF-ADHESIVE SHEET

Clean all surfaces using Armaflex cleaner to remove grease, oil, dirt etc. and cut sheets to size



Peel back release paper & line up sheet. Press firmly to activate adhesive.



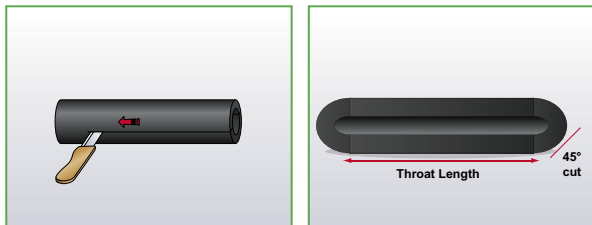
Align material and continue to correctly line up, pressing firmly whilst slowly removing release paper. At butt joints allow 5mm overlap for compression

For additional information and more details please see application guide "Armaflex on ductwork"

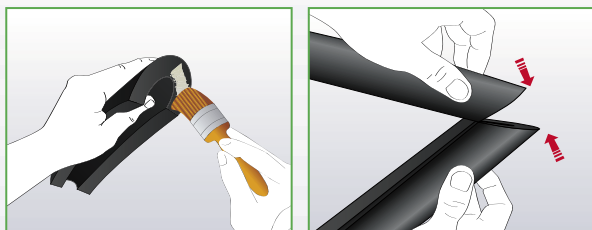
INSULATING DUCT BRACKETS WITH ARMAFLEX

Insulating duct brackets by using Armaflex tubes

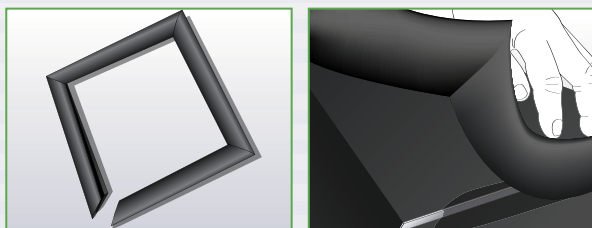
For a cost effective solution with a high-end finish, Armaflex tube can be used to over-cover raised ductwork-connecting brackets.



1. Using unslit Armaflex tube with equal thickness as the attached main duct branch; split with a sharp knife the tube into two equal halves.
2. Measure the four insulated sides of the duct body.
3. Using a mitre box, or the Armaflex template, cut the Armaflex as shown with a 45 degree angle. From the throat measurement, determine the length of the fitting and cut an opposite 45 angle as shown.
4. Continue to cut the other 3 sides of the tube picture frame fitting.

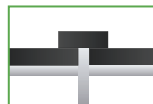


5. Using Armaflex adhesive, apply a thin even film with a brush to the three sets of 45 angles.
6. Allow the adhesive to touch dry, fix the angles together, applying firm even pressure for a good bond.

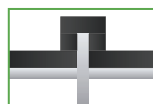


7. Place the picture frame Armaflex tube around the ducting, apply the adhesive to the final angle cuts and bond to complete the fixing of the picture frame.
8. To finish, wet seal around the picture frame fitting cover.

Insulating duct brackets by using Armaflex sheet



Single strip method - Four single strips applied to the insulation.



Three-sided box method - Built up side strips with over-covering body strips.



Continuous single strip method - Complete single continuous strip.

In all situations the fabricated Armaflex insulation bracket covers have the same thickness as the attached ductwork connections. For a continuous vapour sealed system, all insulation covers shall be securely fixed and wet sealed with Armaflex adhesive.

INSULATING CIRCULAR DUCTS WITH ARMAFLEX SHEET

For circular ducts proceed as described in section "Insulating large pipes with Armaflex sheets" & "Armaflex Adhesive and its use on pipework > 88.9 MM O.D." on page 15 of this manual.

For additional information and more details please see application guide "Armaflex on ductwork"

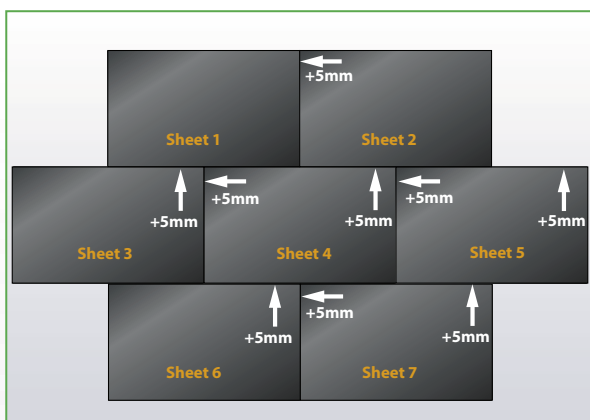
INSULATING VESSELS AND TANKS WITH ARMAFLEX SHEET

Draw up a cutting schedule

Work out the most efficient way of covering the surfaces using Armaflex sheet (2 x 0.5 metre) or continuous roll (1m width and length 4-15m dependent upon the thickness).

Armaflex sheet layout for large vessels and tanks

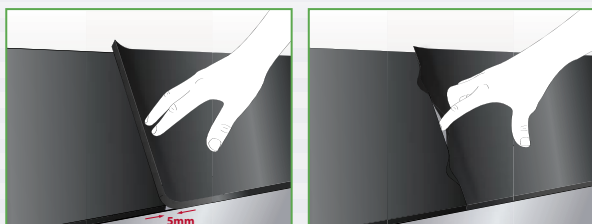
Note: ensure that sheet joints are staggered.



Compression joints

Provide an allowance of an additional 5mm on all dimensions when cutting from Armaflex sheet or roll.

Always make compression joints. On curved surfaces, measure the circumference with a strip of Armaflex of the same thickness to be used for the insulation, including any surface finish. Do not stretch the strip.



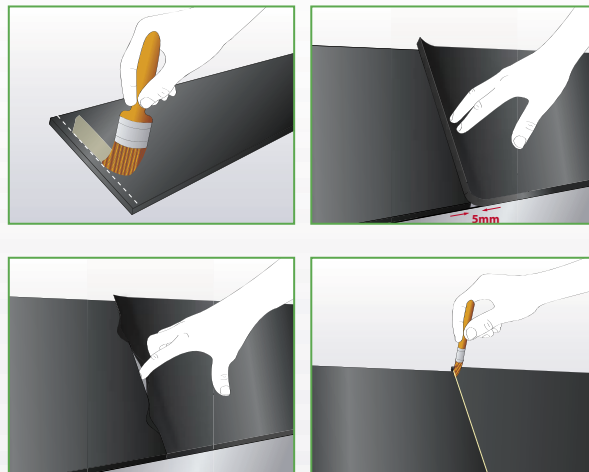
Adhesion

Apply Armaflex adhesive first to the insulation before applying adhesive to the metal surface.

Note: All over adhesive is required on both contacting surfaces in all situations. No "spot" or "strip" adhesive is recommended.

All seams are joined wet. Leave about 15 mm without an adhesive coating on the Armaflex surface. Attach the connecting sheet with adhesive and with 5 mm overlap. Then press in the overlapping butt joint to give additional compression.

Wet sealing of joints on flat surfaces:



Multi-layer application

Where multi-layer insulation is installed use Armaflex cleaner, after 36 hours (see page 10) to remove any talc, chalk, dirt, grease and moisture from the surfaces to be joined. Stagger all seams and butt joints on the second layer relative to the first layer.

Complex shapes

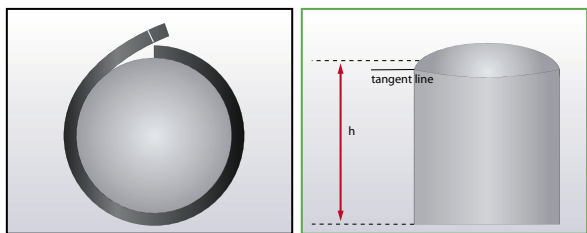
Where complex shapes are to be insulated, the shape of the body is outlined with chalk and this may be transferred directly to the Armaflex sheet by pressing the flexible material against the metal surface so that the chalk is transferred. Cut along the chalked line with a sharp knife to obtain a good fit for the Armaflex.

Outdoor installations

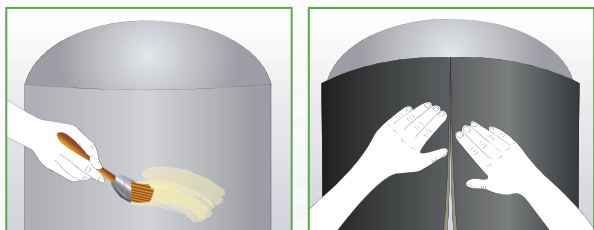
All Armaflex materials (excl. HT/Armaflex) used outdoors will require additional protection against UV radiation. We recommend the use of Armafinish FR paint or one of the Arma-Chek Systems (see page 5).

HT/Armaflex insulation does not require painting when used outdoors.

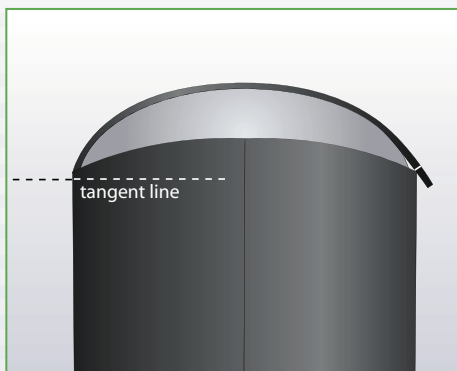
INSULATING SMALL ($\varnothing < 1.5\text{m}$) TANKS AND VESSELS USING ARMAFLEX SHEET



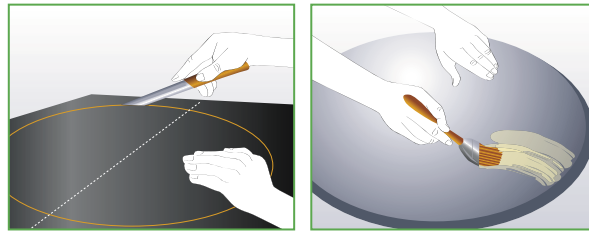
1. Determine the circumference of the tank.
Important: Always measure with a strip of Armaflex of the thickness to be used for the insulation.
Warning: Do not stretch the strip.



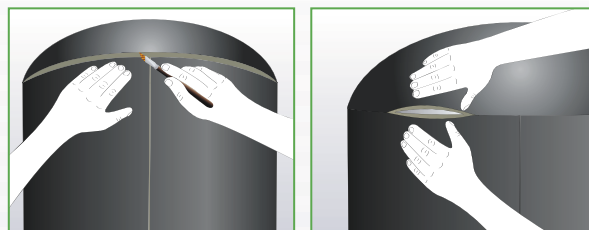
2. Transfer the circumference to the Armaflex sheet and cut to the required size. Spread a thin film of adhesive onto the Armaflex sheet... and then onto the metal surface. When the adhesive is tack dry (fingernail test) place Armaflex sheet in position and press firmly to achieve a good bond.



3. Determine the curve length of the domed surface.
Important: Always measure with a strip of Armaflex of the thickness to be used for the insulation.
Warning: Do not stretch the strip.

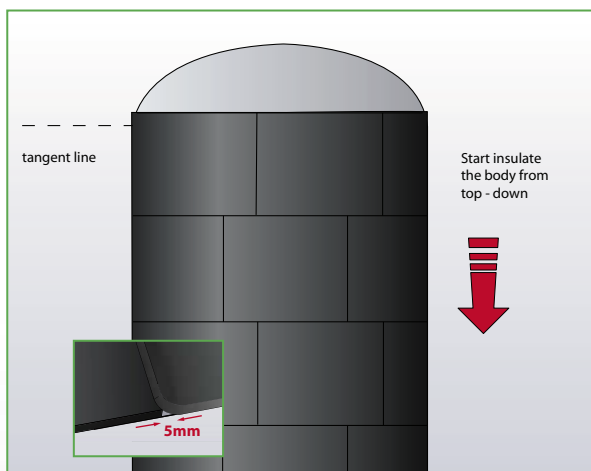


4. Using the curve length as the diameter mark out a complete disc on a piece of Armaflex sheet. If this disc is too large to fit on a single sheet of Armaflex first adhere multiple sheets together.
5. Cut around the disc.
6. Spread a thin film of adhesive onto the back of the Armaflex sheet... and then onto the metal surface.



7. When the adhesive is tack dry (fingernail test) place Armaflex sheet in position onto the top of the tank and press firmly down from the centre, avoiding any slipping of the material, to achieve a good bond.
8. Apply an additional wet seal along the edges at the top of the tank.
9. After the adhesive has been given time to tack dry press the edges of the seam firmly together.

KEY BASIC APPLICATION PROCEDURES FOR LARGE VESSELS Ø > 1.5 MTS.



Install the Armaflex sheet panels from the domed end section of the vessel as shown. Use all-over adhesive coverage to both surfaces.

Continue to install the panels of Armaflex around this domed end section. Ensure the Armaflex sheet is fitted under compression. After the first ring of panels are installed, continue to apply other panels of Armaflex sheet around the body of the vessel as shown.

When the entire surface area of the vessel body is completed, to finish, insulate the dome ends as shown by installing vertical Armaflex panels.

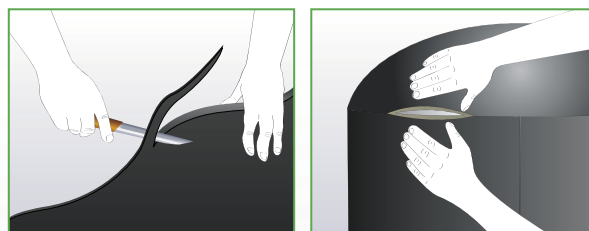


To determine the circular edge cut profile, which is located around the completed body section of dome end – mark the Armaflex circular edge with White marking chalk.

With the first pre-cut Armaflex sheet panel as required, place with a 50mm overrun, the sheet over the profile edge, apply firm pressure and remove. The underside of the Armaflex sheet will show the impression of the required profile.

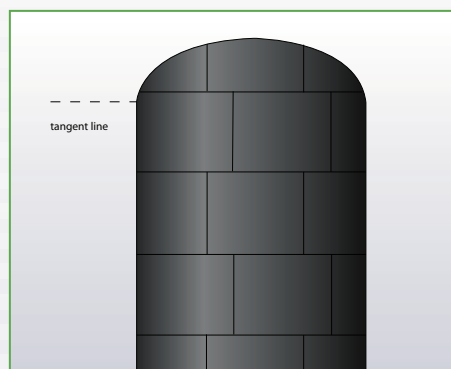
Cut the marked profile with a shape knife and install directly to the section required with all over adhesive coverage. Note: Do not apply adhesive to the circular profile edges of the domed end. This is done last by wet sealing the *jointing details as described below.

Continue to install the required insulation panels as required to complete the domed section.



To finish, *wet seal the domed panels to the main insulated body sections as shown within the section relating to vessels below 1.5 mts.

If required the dome end profile sections can be chamfered, to fit the body panel edge perfectly.



USE OF ADHESIVE IN APPLICATIONS INVOLVING HOT AND HUMID ENVIRONMENTS

High atmospheric humidity's and temperatures lead to faster evaporation of the solvent in Armaflex adhesive. This means that a film of moisture may appear on the surface of the adhesive. Consequently the reliability of the adhesive seam cannot assured as the surfaces to be joined may not bond together.

Under these conditions, the following points may be observed as an alternative to our installation instructions:

- Apply Armaflex adhesive as normal in a thin uniform film on both surfaces.
- Unlike normal bonding, the surfaces to be glued should be held together under pressure whilst wet.

Note: Due to the shorter curing time adhesive can only be applied to a limited area at one time. Depending on the atmospheric humidity, temperature, material thickness and practical installation condition we recommend a tube length of around 1 m as a reference figure.

- To prevent possible tensions within the material and the enclosed solvent opening the seam, seams should be held in place immediately after bonding with Armaflex adhesive tape crosswise to the glued seam every 20 cm or so.

INSULATION OF NON-ARMAFLEX-SYSTEM PIPE SUPPORTS

Armafix pipe supports represent the best option to create a full water vapour sealed system and prevent condensation on cold applications.

Often, cold clamps made of PUR/PIR complete the pipe hanging systems.

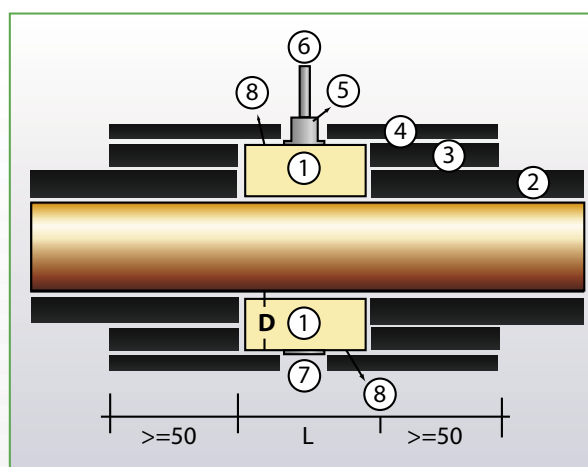
In these cases it is important, that a vapour tight bonding between the cold clamps and the Armaflex insulation is achieved.

This interface represents a thermal bridge where condensation may occur and adhesion at these points requires special attention:

1. Clean the surface of the clamp, using Armaflex cleaner.
2. Apply Armaflex Adhesive on the surfaces which are to be adhered. Allow this first layer of Armaflex adhesive to dry.
3. Apply a second thin coat of Adhesive evenly on both the surface of the clamp and the Armaflex adhesive joints. After the adhesive has cured, the joints should be pressed together in a brief but firm action.
4. When adhering both butt joints under pressure per form a "wet-seal" around the joint, when the clamps have been adhered in advance (please see above)

5. If necessary double the Armaflex thickness to the diameter of the PUR-support.
6. To secure the butt joints apply a overlapping strip of Armaflex using all over adhesive.

Schematic cross section of a connection of Armaflex tubes with a clamp made of PU rigid foam



- ① Clamp made of PU rigid foam
- ② Insulation thickness
D = 20, 30, 40, 50 mm
L = 100 and 250 mm
- ③ Armaflex tube
- ④ Armaflex double layer
- ⑤ Armaflex overlap (thickness ≥ 9 mm)
- ⑥ Connecting thread M10
- ⑦ Threaded bar M10
- ⑧ Two-part screw pipe clamp, zinc coated aluminium facing

REFERENCES

In addition to this manual Armacell provides the following documents, freely available from www.armacell.com/uk (or as part of our ArmaPlus CD). These documents contain further detailed advice for specific applications.

MECHANICAL PROTECTION AND OUTDOOR APPLICATIONS WITH ARMAFLEX

Explanation of the issues arising when installing Armaflex outdoors and evaluation of solutions to protect Armaflex from mechanical damage.

New INSULATING COLD LINES WITH LINE TEMPERATURES BETWEEN -50°C TO -196°C (-58°F to -320°F)

Installation advice on issues arising when insulating low temperature lines below -40°C.

ARMAFLEX UNDERGROUND

Explanation of the theory underlying insulating underground pipes, including advice on insulating underground pipes using Armaflex and also including a calculation tool to calculate the impact of insulation on the time until pipe freezing occurs.

TRACE HEATING

Explanation of the theory underlying insulating trace heated pipes, including advice on selecting correctly dimensioned insulation tube.

ARMAFLEX ON RECTANGULAR & CIRCULAR DUCTWORK

Additional detailed installation advice when installing Armaflex onto rectangular or circular ductwork.

ARMAFLEX IN CONCRETE

Installation advice when burying pipes insulated in Armaflex directly in concrete.

GLUING ARMAFLEX ONTO CELLULAR GLASS

Installation advice when installing Armaflex directly onto a cellular glass surface.

OTHER APPLICATION GUIDES

New Application Guides for Arma-Chek S Plus, Arma-Chek T, Arma-Chek D and Arma-Chek R

- » Special Application Advice for NH/Armaflex
- » Special Application Advice for HT/Armaflex
- » Application Hints for Armaflex Underground
- » Application hints for Armaflex TuffCoat
- » Application of Armaflex DuoSolar VA

New Armaflex application guide for plastic pipes

New Armaflex Protect R-90 application guide

New Application guide for Armaflex on ductwork

- » Application guide for ArmaSound Industrial Systems
- » Armaflex & Arma-Chek application video

CALCULATION TOOLS

» Armafinish FR paint - Coverage Calculator

New Arma-Chek T - Coverage Calculator

» keytec. ISO 15665
Determine the right ArmaSound Industrial Systems

» keytec. Armaflex Underground
Calculate the impact of insulation on the time until pipe freezing occurs.

New keytec. Arma-Chek R fishtail calculator
Calculate the exact shape and measurements for fishtails used to cover bends insulated with Armaflex

New keytec. Unit converter metric / imperial
Calculate the most common units from metric to imperial measurements

» ArmWin AS
Armwin AS is the technical calculation program to determine insulation thicknesses required to prevent surface condensation and limit energy losses. It also allows users to calculate U-values, heat flows and temperature changes for pipes, ducts and tanks.

ARMAFLEX PRODUCTS

CLASS O ARMAFLEX

Closed cell, elastomeric, nitrile rubber insulation material with a Class O fire rating and excellent thermal properties. Available in tubes, pre-slit tubes, sheets, self-adhesive sheets and tapes.

CLASS O ARMAFLEX 15M COILS

Continuous and coils of Class O Armaflex tube ideal for new pipework.

CLASS O ARMAFLEX SELFSEAL

Pre-slit Class O Armaflex tubes with a pair of self adhesive strips down the longitudinal seam. Specifically designed to reduce installation time on existing pipework.

ARMAFLEX AC COILS

Long lengths of continuous Armaflex coils for air conditioning and domestic heating pipes. Achieves a Class O fire rating. Supplied in easy to carry boxes.

HT/ARMAFLEX

Naturally UV resistant closed cell EPDM rubber based Armaflex insulation material capable of operating at line temperatures up to 150°C.

NH/ARMAFLEX

Halogen free, closed cell nitrile rubber based Armaflex insulation material with a low smoke toxicity rating. Achieves a number of maritime fire performance certificates.

CLASS O ARMAFLEX DUCT

The highly flexible, aluminium foil covered Class O Armaflex Duct is designed for insulation of both, rectangular and circular ductwork - matching contours exactly.

ARMAFLEX TUFFCOAT

Class O Armaflex tubes with a tough white covering pre-applied. Covering provides protection against UV exposure, impact damage and weathering.

ARMAFLEX SPLIT / DUOSPLIT

Copper pipes pre-insulated with a naturally UV resistant closed cell EPDM rubber based Armaflex insulation. Insulation comes with a tough white covering pre-applied. Designed for use with split air conditioning systems.

ARMAFLEX DUOSOLAR

Parallel stainless steel or copper pipes pre-insulated with a naturally UV resistant closed cell EPDM rubber based Armaflex insulation. Insulation comes with a tough black covering pre-applied. Designed to withstand the temperatures associated with solar hot water systems.

ARMAFLEX PROTECT R-90

Effective fire penetration sealing for pipe and ductwork. Available in tubes and sheet, Armaflex Protect R-90 provides up to 120 minutes insulation and integrity ratings.

ARMAFIX PIPE SUPPORT

Armaflex sections with load bearing PUR/PIR inserts and an aluminium outer cladding to prevent excessive material compression.

ARMALOAD PIPE SUPPORT & SECTIONS

High density sections of Armaflex designed to resist material compression.

ARMAFLEX ACCESSORIES

Armaflex Adhesive 520, for adhering nitrile rubber based Armaflex materials and Armaflex Adhesive 625 for adhering EPDM based Armaflex materials.

Armafinish FR paint, for visual impact and to prevent damage from UV exposure when Armaflex is used outside.



The Armaflex Assurance System

The Armaflex Assurance System - your peace-of-mind assurance that Armaflex is made to the strictest quality, technical performance and environmentally friendly standards.

Prevents Moisture Permeation

Multi-layered closed wall cells provide a built-in vapour barrier which can't be compromised by surface punctures or rips - this ensures the vapour barrier stays working for the lifetime of the insulation system.

Provides Long-Term Thermal Performance

Non-wicking - No transmission of water vapour is possible through the insulation due to the closed cell nature of Armaflex. Moisture will not migrate through professionally installed Armaflex.

Long-Term Durability

No fragile water vapour barrier - The built-in vapour barrier can't be "damaged" during installation or over the lifetime of the insulation.

Avoids Thermal Bridging

Armaflex is a seamless insulation system - these same technical properties provide a secure system.

Long-Term Energy Efficiency

By minimising the ingress of water vapour into the insulation the multi-layered cells in Armaflex prevent the thermal conductivity from rising. Increased thermal conductivity can in the long term lead to increased energy costs and surface condensation.

Inhibits Growth of Mould

- Minimises moisture permeation, the first requirement for mould growth
- Minimises the second requirement: No cellulosic vapour jacket, and the smooth surface doesn't trap dust

Formaldehyde-Free

Ensures that Armaflex is a low-emitting product. It is also low VOC.

Support Requirements of Health & IAQ (Indoor-Air-Quality)

- Dust and Fibre-Free
- Will not emit any particles or fibres - even when cut into shapes and fittings
- No outgassing - CFC & HCFC free manufacturing process

Supervised Manufacturing & Technical Values

Factory Mutual Approval Guide - guarantee consistent performance to published values by monitoring

- Fire Performance (FM pipe chase test, Flammability test)
- Water Vapour Transmission
- Thermal Conductivity

Supervised Quality Standards

ISO Registration ensures that all Armacell manufacturing facilities consistently meet customer requirements . . . by managing all processes with ongoing quality assessments.



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